Hidden axial matter in relativistic $\ell = 1$ boson stars: Coupled modes, ringdown, and resonant response

Adel H. Al-Yoorby[1,*]

[1]*Department of Physics, Khalifa University of Science and Technology, P.O. Box 127788, Abu Dhabi, United Arab Emirates*

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Abstract

We identify a relativistic odd-parity matter channel that is absent from ordinary single-field boson stars. For the $\ell = 1$, total-angular-momentum $J = 2$, orbital $L = 2$ scalar perturbation, an exact tensor-harmonic reduction gives a vanishing density projection but nonzero axial current and odd anisotropic stress. In normalized harmonics and a fixed Condon–Shortley phase convention, $\langle X^A, S_A \rangle = -\mathrm{i}/(2\sqrt{\pi})$ and $\langle X^{AB}, Q_{AB} \rangle = \mathrm{i}/\sqrt{\pi}$. The result survives the sum over all three internal magnetic states, is independent of $M = -2, \ldots, 2$, and vanishes under the corresponding axial projection for the ordinary $\ell = 0$ scalar. We derive the closed six-state axial Einstein–Klein–Gordon system, propagate its gravitational and massive-scalar channels through the Schwarzschild exterior, and impose a two-sided complex-frequency match. A declared 7×5 complex-frequency search discovers one pole inside a subsequently counted contour. Seven independently solved stable backgrounds then continue the same counted branch over $0.0953 \leq C_{99} \leq 0.1131$, with adjacent profile overlaps exceeding 0.9991. For the $a_1^0 = 0.08$ star, two refined domains give $\sigma \simeq 0.0493977307 - 5.96519 \times 10^{-7}\mathrm{i}$ and $Q \simeq 4.1405 \times 10^4$; a predeclared targeted fixed-grid real-frequency Lorentzian, fitted without loading the stored pole, reproduces the width within 4.5×10^{-5} fractionally using the same operator. A short 1 + 1 evolution checks preservation of the oscillation frequency but is not long enough to measure the damping time. We report the raw two-channel conversion amplitude and the explicitly normalized stationary axial response $B/(AM^5) = -122.47418$, while separating the still-missing canonical conversion flux from the provisional pole claim. The central result is that internal scalar angular motion, invisible to the Newtonian density channel, produces strong evidence for a long-lived relativistically radiating axial mode in a geometrically spherical star.

Keywords: ℓ-boson stars, axial perturbations, odd parity, quasinormal modes, resonant response

I. INTRODUCTION

Boson stars are self-gravitating configurations of complex scalar fields whose macroscopic support is provided by harmonic time dependence rather than a material pressure law. They provide a controlled setting for studying compact, horizonless relativistic objects, scalar-field dynamics, gravitational-wave phenomenology, and alternatives to black holes. The ℓ-boson-star construction is especially useful because a multiplet of $2\ell + 1$ fields can carry nontrivial angular structure while their

* alyoorby@gmail.com; ORCID: 0009-0000-1935-4392

summed stress tensor remains spherically symmetric [1]. This diagonal spatial–internal symmetry produces equilibrium families that are spherical at the level of the metric but retain internal angular momentum structure in the matter sector.

The radial stability of these configurations was analyzed relativistically by Alcubierre *et al.* [2]. That work derived a coupled pulsation system, identified the change of sign of the fundamental eigenvalue near the maximum-mass turning point, and tabulated a benchmark ground mode and first overtone. Those results supply an unusually stringent code-validation problem: the background normalization, regular center series, coupled perturbation coefficients, physical outer conditions, node classification, and the sign of σ^2 must all be consistent.

The central question of this work is how the matter multiplet's internal orbital structure changes relativistic parity physics. In the nonrelativistic decomposition, coupling the background $\ell = 1$ representation to an orbital harmonic gives $L = J - 1, J, J + 1$; the $L = J$ channel has odd parity and does not perturb the Newtonian scalar potential [3]. General relativity is more discriminating: it couples not only to density but also to angular momentum density, transverse current, and anisotropic stress. Whether those sources survive the sum over the three background fields is therefore a decisive calculation, not a consequence that may be inferred from parity alone.

We perform that calculation explicitly for the first radiative case, $\ell = 1$, $J = L = 2$. The density bilinear cancels exactly, while both the axial vector and axial tensor projections are nonzero and independent of M. We close the radial Einstein–Klein–Gordon system, discover and count its first long-lived complex mode, continue it through seven stable backgrounds, and resolve the central pole under real-frequency driving. A short time evolution supplies a real-frequency consistency check. The boundary-value and channel-matching strategy builds on the established theory of stellar QNMs and coupled-field exterior matching [4, 5].

The contribution is three-tiered. First, we derive and independently verify the new axial angular source, including a Newtonian density-null identity and an $\ell = 0$ control. Second, we derive and solve the closed axial gravito-scalar system with physical channel sheets, derivative-consistent background coefficients, from-scratch root discovery, and an exterior-algebra Evans count. Third, we connect the pole to a predeclared targeted fixed-grid driven resonance whose fit does not load the stored pole, and delimit the current status of time-domain damping and open-channel flux normalization. The equilibrium and radial stability campaign supplies the validated substrate. Charged equilibria and an occupation-noise toy estimate are distributed separately and are not evidence for neutral axial spectroscopy.

The paper is organized as follows. Section II fixes conventions and validation criteria. Section III gives the equilibrium problem. Section IV derives the odd scalar channel, followed by the closed dynamics and response experiments in Sections V–VII. Sections VIII–X certify the radial substrate, and Sec. XI restores physical units. Sections XII and XIII collect the interpretation and reproducibility evidence.

A. Literature context and scope

The original mini-boson-star solutions of Kaup and of Ruffini and Bonazzola established that a massive complex scalar field can form a regular, asymptotically flat, self-gravitating configuration without a material equation of state [6, 7]; self-interactions can change the characteristic mass scale substantially [8]. Their equilibrium, stability, and phenomenology have since been developed through perturbative, variational, catastrophe-theory, and nonlinear-evolution methods; broad reviews are given in Refs. [9–12]. For the ordinary $\ell = 0$ family, the maximum-mass turning point is tied to the appearance of a radial instability [13–17]. Its nonspherical perturbations were formulated in tensor harmonics by Kojima, Yoshida, and Futamase [18], and subsequent QNM calculations connected the stellar spectrum to gravitational-wave phenomenology [5, 19–21]. More generally, relativistic stellar QNMs can be formulated as a scattering problem for incident gravitational waves, with distinct axial and polar sectors [22].

The ℓ-boson-star construction enlarges this setting by populating all magnetic substates of a fixed angular number with equal radial profiles. Although each field carries angular dependence, the addition theorem makes the total stress tensor spherical. The resulting solutions can retain matter-sector angular structure without requiring an axisymmetric metric. Alcubierre *et al.* constructed the equilibrium families and subsequently derived their fully relativistic radial pulsation equations [1, 2]. Their analysis established equilibrium sequences, radial stability, and benchmark eigenfrequencies. Here we extend that foundation to the relativistic gravito–scalar odd sector and certify its first long-lived axial matter mode together with its time-domain, scattering, resonant, and stationary manifestations.

Ordinary boson-star perturbations have been studied with frequency-domain, time-domain, and catastrophe-theory methods. Their scalar perturbations are polar, leaving the axial gravitational equation matter-free at linear order [5, 18, 19]. The collectively spherical $\ell = 1$ background is different: its $L = J$ scalar channel is odd under parity [3]. Section IV shows explicitly that this channel remains matter-carrying in relativity even though its leading density projection vanishes.

Two broader calculations are retained only as context. The electromagnetic calculation solves charged Einstein–Maxwell–Klein–Gordon equilibria but no charged perturbation spectrum [23–26]. The second is a coherent-state occupation-noise toy estimate based on the computed Noether-charge sequence, not a renormalized semiclassical stress-tensor calculation [27, 28]. Neither result is folded into the neutral eigenvalue or used to support the axial discovery.

The distinction matters because agreement between two discretizations is not automatically agreement between two physical formulations. The boundary-value and spectral methods below use different algebraic representations and independently coded radial coefficients, but they share the same equilibrium profiles and published perturbation model. We counter common-formulation risk with literature benchmarks, direct center and matrix-assembly identities, physical boundary checks, independent coefficient transcription, domain continuation, and eigenfunction comparisons.

II. CONVENTIONS AND NUMERICAL ACCEPTANCE

We use signature $(-,+,+,+)$ and units

$$G = c = \hbar = \mu = 1, \tag{1}$$

unless a dimensional factor is shown. The spherically symmetric metric is

$$\mathrm{d}s^2 = -\alpha(r)^2\mathrm{d}t^2 + \gamma(r)^2\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \qquad \gamma^2 = \left(1 - \frac{2M(r)}{r}\right)^{-1}. \tag{2}$$

There are $\kappa_\ell = 2\ell + 1$ equal-mass complex fields,

$$\Phi_{\ell m}(t,r,\theta,\phi) = e^{+\mathrm{i}\omega t}\psi_\ell(r)Y_{\ell m}(\theta,\phi), \qquad \sum_{m=-\ell}^{\ell} |Y_{\ell m}|^2 = \frac{2\ell+1}{4\pi}. \tag{3}$$

Perturbations use $e^{-\mathrm{i}\sigma t}$, so $\mathrm{Im}\,\sigma < 0$ is damped and $\mathrm{Im}\,\sigma > 0$ is unstable. The radial problem is real and depends on σ^2; negative σ^2 therefore represents exponential growth without changing the numerical code path.

Every headline numerical result must survive five classes of checks: outer-domain variation, discretization variation, branch or mode continuity, direct equation and boundary diagnostics, and an independent numerical method. For radial modes, node count and normalized physical-eigenfunction overlap are part of mode identification. Deterministic numerical shifts are not assumed statistically independent; their conservative sum is the headline uncertainty, while quadrature is retained only as a secondary descriptive quantity.

TABLE I. Frozen conventions and their numerical consequences.

Quantity	Convention	Consequence
Geometry	Areal radius and signature $(-,+,+,+)$	$\gamma^2 = (1-2M/r)^{-1}$
Matter	$2\ell+1$ equal-mass complex fields	$\kappa_\ell = 2\ell+1$
Center amplitude	$\psi_\ell/r^\ell \to a_\ell^0/\kappa_\ell$	Reproduces the published background normalization
Perturbation time	$e^{-\mathrm{i}\sigma t}$	$\mathrm{Im}\,\sigma < 0$ denotes damping
Radius	$M(R_{99}) = 0.99M_T$	R_{999} is stored and reported separately
Validation	Domain, resolution, method, residual, and shape tests	Solver success alone is insufficient

III. RELATIVISTIC $\ell = 1$ BACKGROUNDS

A. Action, multiplet ansatz, and spherical stress tensor

We begin with $\kappa_\ell = 2\ell+1$ identical complex scalar fields of mass μ and action

$$S = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}\sum_{m=-\ell}^{\ell}\left(g^{ab}\nabla_a\Phi_{\ell m}^*\nabla_b\Phi_{\ell m} + \mu^2|\Phi_{\ell m}|^2\right)\right]. \tag{4}$$

Variation with respect to the metric and conjugate fields gives

$$G_{ab} = 8\pi G\,T_{ab}, \qquad \left(\Box - \mu^2\right)\Phi_{\ell m} = 0, \tag{5}$$

where

$$\begin{aligned} T_{ab} = \frac{1}{2}\sum_{m=-\ell}^{\ell}\Big[&\nabla_a\Phi_{\ell m}^*\nabla_b\Phi_{\ell m} + \nabla_b\Phi_{\ell m}^*\nabla_a\Phi_{\ell m} \\ &- g_{ab}\left(\nabla^c\Phi_{\ell m}^*\nabla_c\Phi_{\ell m} + \mu^2|\Phi_{\ell m}|^2\right)\Big]. \end{aligned} \tag{6}$$

For the harmonic multiplet

$$\Phi_{\ell m} = e^{\mathrm{i}\omega t}\psi_\ell(r)Y_{\ell m}(\theta,\phi), \tag{7}$$

the spherical-harmonic addition identities imply

$$\sum_{m=-\ell}^{\ell} Y_{\ell m}^* Y_{\ell m} = \frac{\kappa_\ell}{4\pi}, \qquad \sum_{m=-\ell}^{\ell}\nabla_A Y_{\ell m}^*\nabla_B Y_{\ell m} = \frac{\ell(\ell+1)\kappa_\ell}{8\pi}\Omega_{AB}. \tag{8}$$

Consequently, the total stress tensor is spherical even though each member of the multiplet carries angular dependence. The scalar fields are time dependent, but their common harmonic phase cancels in all stress-energy bilinears, leaving a static geometry.

B. Einstein–Klein–Gordon reduction

For $\kappa = 2\ell + 1$, the static equations in the normalization of Ref. [2] are

$$M' = \frac{\kappa r^2}{2}\left[\frac{\psi'^2}{\gamma^2} + \left(\frac{\omega^2}{\alpha^2} + 1 + \frac{\ell(\ell+1)}{r^2}\right)\psi^2\right], \tag{9}$$

$$\frac{\alpha'}{\alpha} = \gamma^2\left\{\frac{M}{r^2} + \frac{\kappa r}{2}\left[\frac{\psi'^2}{\gamma^2} + \left(\frac{\omega^2}{\alpha^2} - 1 - \frac{\ell(\ell+1)}{r^2}\right)\psi^2\right]\right\}, \tag{10}$$

$$\psi'' + \left(\frac{2}{r} + \frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)\psi' + \gamma^2\left(\frac{\omega^2}{\alpha^2} - 1 - \frac{\ell(\ell+1)}{r^2}\right)\psi = 0. \tag{11}$$

The radial metric derivative is evaluated from the mass definition,

$$\frac{\gamma'}{\gamma} = \gamma^2\left(\frac{M'}{r} - \frac{M}{r^2}\right), \tag{12}$$

rather than by numerically differentiating an interpolated metric profile.

C. Regular center expansion

Regularity and analyticity require

$$\psi_\ell(r) = \frac{a_\ell^0}{\kappa_\ell} r^\ell + O(r^{\ell+2}), \qquad M(r) = O(r^{2\ell+1}), \tag{13}$$

and $\alpha(r) = \alpha_c + O(r^2)$. For $\ell = 1$, let

$$A \equiv \lim_{r\to 0}\frac{\psi_1(r)}{r} = \frac{a_1^0}{3}. \tag{14}$$

Substitution in Eq. (9) gives

$$M(r) = \frac{3A^2}{2} r^3 + O(r^5), \tag{15}$$

while $\psi_1(r) = Ar + O(r^3)$ and $\psi_1'(r) = A + O(r^2)$. These leading terms are imposed at a small excision radius. The field-equation residual is then evaluated as the excision radius is varied, which distinguishes a valid Frobenius start from a merely finite numerical start.

D. Bound-state asymptotics and outer boundary

Asymptotic flatness requires

$$M(r) \to M_T, \qquad \alpha(r) \to \sqrt{1 - \frac{2M_T}{r}}, \qquad r \to \infty. \tag{16}$$

Because the scalar is massive, localization further requires $0 < \omega < 1$. Defining

$$k = \sqrt{1 - \omega^2}, \tag{17}$$

the first curvature-corrected tail is

$$\psi_\ell(r) \sim r^{-p} e^{-kr}, \qquad p = 1 + \frac{M_T(1 - 2\omega^2)}{k}. \tag{18}$$

At the finite numerical boundary this becomes

$$\psi'(r_{\text{out}}) + \left(k + \frac{p}{r_{\text{out}}}\right)\psi(r_{\text{out}}) = 0. \tag{19}$$

The frequency is represented by an unconstrained parameter passed through a logistic map, which keeps every nonlinear iterate inside the bound-state interval.

E. Global observables and radius conventions

The total Noether charge and the mass-fraction radii are

$$Q = \kappa\omega \int_0^\infty \frac{\psi^2\gamma}{\alpha} r^2 \mathrm{d}r, \qquad M(R_p) = pM_T, \qquad M_T = \lim_{r\to\infty} M(r). \tag{20}$$

The associated compactness and binding diagnostic are

$$C_p = \frac{M_T}{R_p}, \qquad E_{\text{bind}} = M_T - Q. \tag{21}$$

We store both $p = 0.99$ and $p = 0.999$ because the published maximum-mass radius is numerically closer to the latter. The sign of E_{bind} is useful energetically, but the radial eigenvalue provides the direct dynamical stability test.

F. Nonlinear collocation and continuation

The background equations are solved as a nonlinear two-point boundary-value problem with adaptive collocation on a geometric radial mesh, using the collocation implementation documented with SciPy [29]. The unknown frequency is included as a solver parameter. Continuation starts from the validated $a_1^0 = 0.08$ configuration and advances along the nodeless branch in steps no larger than 0.005. Dense output is used to compute global charges and mass-fraction radii, while independent first-order residuals and the scalar-tail logarithmic derivative diagnose the stored solution.

The production settings are $r_{\max} = 80$, 800 initial mesh points, relative collocation tolerance 10^{-7}, and at most 40,000 adaptive nodes. Outer-domain checks use $r_{\max} = 60, 80, 100$ at otherwise fixed settings. Independent DOP853 re-integrations in Sec. X C 2 then test propagation of the collocation profiles with a distinct algorithm.

The initial geometric mesh resolves the regular center and the exponential tail without compactifying infinity. A converged model at one central amplitude is interpolated to initialize the next amplitude, with separate upward and downward continuations from the $a_1^0 = 0.08$ anchor. Each stored solution is subsequently evaluated on a dense geometric grid. On that grid we recompute Eqs. (9)–(11), the metric identity (12), and the Robin condition (19). These post-solve diagnostics are independent of the collocation residual returned internally by the nonlinear algorithm.

IV. ODD SCALAR CHANNEL AND AXIAL STRESS TENSOR

A. Diagonal angular momentum and parity

The spherical metric does not imply that every scalar perturbation is polar. Spatial rotations act simultaneously on the orbital harmonics and on the internal multiplet. It is therefore convenient to make a unitary change of internal basis and couple an orbital representation L to the background representation ℓ. With $s = -\ell, \ldots, \ell$, define the normalized background angular vector and coupled perturbation harmonic

$$B_s^{(\ell)} = \frac{(-1)^\ell}{\sqrt{2\ell + 1}} Y_{\ell s}^*, \tag{22}$$

$$P_s^{JML} = \sum_{m_L=-L}^{L} \langle L m_L, \ell s | J M \rangle Y_{L m_L}. \tag{23}$$

The radial field multiplying Eq. (22) is $F = \sqrt{2\ell + 1}\, \psi_\ell$, so this normalized basis leaves the physical sum over fields unchanged. The allowed values satisfy $|J - \ell| \le L \le J + \ell$. For $\ell = 1$, the $L = J - 1, J + 1$ channels are even and the $L = J$ channel is odd [3].

The scalar density bilinear relevant at leading nonrelativistic order is

$$\mathcal{D}^{JML} = \sum_s B_s^* P_s^{JML}. \tag{24}$$

Its reduced coefficient is proportional to $\langle L0, \ell 0 | J0 \rangle$. For $\ell = 1$ and $J = L = 2$, the sum $J + L + \ell = 5$ is odd; hence

$$\mathcal{D}^{2M2} = 0 \qquad (M = -2, -1, 0, 1, 2). \tag{25}$$

This is the group-theoretic origin of the density-silent odd sector in the Schrödinger–Poisson limit. It does not imply that every component of the relativistic stress tensor vanishes.

B. Axial vector and tensor projections

Let D_A and ε_{AB} be the covariant derivative and volume form of the unit two-sphere. We use the normalized axial vector harmonic and its symmetrized derivative,

$$X_A^{JM} = -\frac{\varepsilon_A{}^B D_B Y_{JM}}{\sqrt{J(J+1)}}, \qquad X_{AB}^{JM} = D_{(A} X_{B)}^{JM}. \tag{26}$$

Our normalization gives

$$\int X_A^* X^A \, \mathrm{d}\Omega = 1, \qquad \int X_{AB}^* X^{AB} \, \mathrm{d}\Omega = \frac{(J-1)(J+2)}{2}. \tag{27}$$

The angular structures entering the mixed and trace-free angular components of the scalar stress tensor are

$$S_A^{JML} = \sum_s B_s^* D_A P_s^{JML}, \qquad Q_{AB}^{JML} = \sum_s D_{(A} B_s^* D_{B)} P_s^{JML}. \tag{28}$$

Direct evaluation for $\ell = 1$, $J = L = 2$ yields the pointwise identities

$$S_A^{2M2} = -\frac{\mathrm{i}}{2\sqrt{\pi}} X_A^{2M}, \qquad Q_{AB}^{2M2} = \frac{\mathrm{i}}{2\sqrt{\pi}} X_{AB}^{2M}. \tag{29}$$

Consequently,

$$\int X^{A*} S_A \, \mathrm{d}\Omega = -\frac{\mathrm{i}}{2\sqrt{\pi}}, \qquad \int X^{AB*} Q_{AB} \, \mathrm{d}\Omega = \frac{\mathrm{i}}{\sqrt{\pi}}. \tag{30}$$

The factor of two between the second pointwise and integrated coefficients is the $J = 2$ tensor-harmonic norm in Eq. (27). Overall phases change if the tensor-harmonic basis is rephased, but vanishing, magnitudes, and M degeneracy do not.

The $M = 0$ result was also obtained symbolically from explicit low-order spherical harmonics, without numerical quadrature. The numerical calculation uses Gauss–Legendre integration in $\cos\theta$ and uniform Fourier nodes in ϕ. Increasing the grid from 24×32 to 48×64 changes each projected coefficient by less than 2×10^{-12}; at the reported resolution the spread over M is 2.9×10^{-15} for the vector and 5.8×10^{-15} for the tensor projection.

TABLE II. Independent quadrature check of the $\ell = 1$, $J = L = 2$ angular identities. The exact values are $|\langle X, S\rangle| = 1/(2\sqrt{\pi}) = 0.2820947917738781$ and $|\langle X_{(2)}, Q\rangle| = 1/\sqrt{\pi} = 0.5641895835477563$.

M	$\|\mathcal{D}\|_2$	$\mathrm{Im}\langle X, S\rangle$	$\mathrm{Im}\langle X_{(2)}, Q\rangle$
-2	2.83×10^{-17}	-0.2820947917738770	0.5641895835477598
-1	2.68×10^{-17}	-0.2820947917738799	0.5641895835477542
0	1.01×10^{-17}	-0.2820947917738773	0.5641895835477541
1	2.69×10^{-17}	-0.2820947917738799	0.5641895835477542
2	2.83×10^{-17}	-0.2820947917738770	0.5641895835477598

C. Linearized relativistic source

To expose the physical content, consider one frequency sideband in the same time convention as the background,

$$\Phi_s^{(0)} = e^{\mathrm{i}\omega t}F(r)B_s, \qquad \delta\Phi_s^{(+)} = e^{\mathrm{i}\omega t-\mathrm{i}\sigma t}u(r)P_s^{2M2}. \tag{31}$$

The companion sideband required by the full complex perturbation contributes the conjugate angular structure with an independent radial amplitude. For the sideband in Eq. (31), Eq. (25) implies $\sum_s P_s D_A B_s^* = -S_A$. Substitution into the stress tensor (6) gives

$$\delta T_{tA}^{(+,\mathrm{scalar})} = -\frac{\mathrm{i}}{2}F(2\omega-\sigma)uS_A, \tag{32}$$

$$\delta T_{rA}^{(+,\mathrm{scalar})} = \frac{1}{2}(F'u - Fu')S_A, \tag{33}$$

$$\delta T_{AB}^{(+,\mathrm{scalar,odd})} = FuQ_{AB}. \tag{34}$$

Using Eq. (29), the projected sources are therefore

$$\delta T_{tA}^{(+,\mathrm{scalar})} = -\frac{F(2\omega-\sigma)u}{4\sqrt{\pi}}X_A^{2M}, \tag{35}$$

$$\delta T_{rA}^{(+,\mathrm{scalar})} = -\frac{\mathrm{i}(F'u - Fu')}{4\sqrt{\pi}}X_A^{2M}, \tag{36}$$

$$\delta T_{AB}^{(+,\mathrm{scalar,odd})} = \frac{\mathrm{i}Fu}{2\sqrt{\pi}}X_{AB}^{2M}. \tag{37}$$

These equations demonstrate that the cancellation of the density channel does not remove the relativistic angular current or odd anisotropic stress.

For completeness, write the odd metric perturbation as

$$h_{tA} = h_0 X^{2M}_A, \qquad h_{rA} = h_1 X^{2M}_A, \qquad h_{AB} = h_2 X^{2M}_{AB}. \tag{38}$$

If

$$\mathcal{K}_0 = \sum_s \left(\nabla^c \Phi^{(0)*}_s \nabla_c \Phi^{(0)}_s + \mu^2 |\Phi^{(0)}_s|^2\right), \tag{39}$$

then variation of the explicit metric in Eq. (6) adds $-\mathcal{K}_0 h_{ab}/2$. Terms proportional to the background g_{AB} are pure angular trace and have zero axial-tensor projection. Thus the complete axial projections for this sideband are obtained from Eqs. (35)–(37) by adding $-\mathcal{K}_0 h_0 X_A/2$, $-\mathcal{K}_0 h_1 X_A/2$, and $-\mathcal{K}_0 h_2 X_{AB}/2$, respectively. Regge–Wheeler gauge sets $h_2 = 0$ for $J \geq 2$. The angular system therefore closes on the usual axial metric harmonics, but with a genuine scalar source.

D. Null control and weak-field interpretation

For an ordinary $\ell = 0$ background, $B = Y_{00}$ is constant. The vector bilinear is then proportional to $D_A Y_{JM}$ and is purely polar, while $Q_{AB} = 0$. Both axial projections vanish exactly. This control shows that Eq. (29) is not an artifact of projecting an arbitrary scalar gradient: it requires the internal orbital structure of the $\ell = 1$ background.

In the nonrelativistic limit the leading mass-density perturbation is proportional to Eq. (24), so the odd sector does not source the Schrödinger–Poisson potential, in agreement with Ref. [3]. The relativistic current and anisotropic-stress terms are nevertheless nonzero. Their metric response becomes post-Newtonianly suppressed as compactness decreases. The radial axial equations, massive-scalar sheets, and outgoing gravitational boundary conditions developed below quantify that response and resolve both the QNM frequency and its damping law.

E. Verified metric–scalar radial coupling

The first dynamical step beyond the angular proof is to vary the Klein–Gordon operator with respect to the odd metric. In Regge–Wheeler gauge, let the positive sideband have radial amplitude $u_+(r)$ and total time dependence $e^{\mathrm{i}(\omega-\sigma)t}$. Direct coordinate variation of $\Box\Phi$, without using the

Einstein equations or a published component formula, gives

$$0 = \frac{1}{\alpha\gamma r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\alpha r^2}{\gamma}u_+'\right) + \left[\frac{(\omega-\sigma)^2}{\alpha^2} - \mu^2 - \frac{L(L+1)}{r^2}\right]u_+ \\ + \frac{\kappa_V}{r^2}\left[\frac{\mathrm{i}F(2\omega-\sigma)}{\alpha^2}h_0 - \frac{1}{\gamma^2}\left\{Fh_1' + \left(2F' + F\left(\frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)\right)h_1\right\}\right], \tag{40}$$

where $L = 2$ and $\kappa_V = -\mathrm{i}/(2\sqrt{\pi})$ in the convention of Eq. (29). The companion amplitude u_- obeys the same radial differential operator with $(\omega-\sigma)^2$ replaced by $(\omega+\sigma)^2$; its time-current coupling changes to $-\mathrm{i}F(2\omega+\sigma)h_0/\alpha^2$, while the radial h_1 coupling is unchanged in this phase convention.

Equation (40) has been checked by two algebraically independent calculations. The first varies $g^{ab}(\partial_a\partial_b - \Gamma^c{}_{ab})\Phi$ in coordinates on a general spherical background. The second expands the compact expression shown above. Their symbolic difference simplifies identically to zero. Setting $h_0 = h_1 = 0$ recovers the massive scalar sideband equation; setting $F = 0$ removes the scalar–metric coupling.

This calculation also corrects a useful bookkeeping misconception. The two scalar sidebands obey second-order radial equations. Therefore a local first-order formulation cannot close on only (h_0, h_1, u_+, u_-). Before Einstein constraints are used to eliminate variables, the minimal state is six-dimensional, for example

$$\boldsymbol{Y} = (h_0, h_1, u_+, u_+', u_-, u_-')^T. \tag{41}$$

The next section combines Eq. (40) and its companion with two axial Einstein equations, imposes regular central data and outgoing/decaying channel conditions, and performs a declared complex search. It also records which certification gates are closed and which remain open, so the reported frequency is interpreted as a strong candidate pole rather than a completed independent damping measurement.

V. CLOSED NEUTRAL AXIAL DYNAMICS AND FIRST COMPLEX MODE

This section gives the current frequency-domain calculation: the radial Einstein operator, both scalar sidebands, regular-center and radiative exterior data, a two-sided complex-frequency match, and its remaining certification limits. The parity decomposition and vacuum exterior reduce to the Regge–Wheeler framework [30], while the coupled-channel boundary logic follows the direct matching approach used for relativistic stars with interacting fields [5].

A. Directly linearized axial Einstein tensor

We set $h_2 = 0$ and use $e^{-\mathrm{i}\sigma t}$. A direct coordinate variation of the Einstein tensor on the general spherical metric of Eq. (2) gives the following coefficients of X_A and X_{AB} for $J = 2$:

$$\begin{aligned}\delta G_{tA}/X_A = & -\frac{\mathrm{i}\sigma}{2\gamma^2}h_1' + \frac{\mathrm{i}\sigma}{2\gamma^2}\left(\frac{\gamma'}{\gamma} + \frac{\alpha'}{\alpha}\right)h_1 - \frac{h_0''}{2\gamma^2} + \frac{h_0'}{2\gamma^2}\left(\frac{\gamma'}{\gamma} + \frac{\alpha'}{\alpha}\right) \\ & + \frac{h_0}{\gamma^2}\left(\frac{\alpha''}{\alpha} - \frac{\alpha'\gamma'}{\alpha\gamma}\right) - \frac{\mathrm{i}\sigma h_1}{r\gamma^2} - \frac{2\gamma' h_0}{r\gamma^3} + \left(2 + \frac{1}{\gamma^2}\right)\frac{h_0}{r^2},\end{aligned} \tag{42}$$

$$\begin{aligned}\delta G_{rA}/X_A = & -\frac{\sigma^2}{2\alpha^2}h_1 + \frac{\mathrm{i}\sigma}{2\alpha^2}h_0' - \frac{\mathrm{i}\sigma}{r\alpha^2}h_0 \\ & + h_1\left[\frac{\alpha''}{\alpha\gamma^2} - \frac{\alpha'\gamma'}{\alpha\gamma^3} - \frac{\gamma'}{r\gamma^3} + \frac{\alpha'}{r\alpha\gamma^2} + \frac{2}{r^2}\right],\end{aligned} \tag{43}$$

$$\delta G_{AB}/X_{AB} = \frac{\mathrm{i}\sigma}{\alpha^2}h_0 + \frac{1}{\gamma^2}h_1' + \frac{1}{\gamma^2}\left(\frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)h_1. \tag{44}$$

These expressions were obtained by differentiating the exact Christoffel and Ricci formulas with respect to a metric bookkeeping parameter. Their flat-vacuum reduction is a decisive sign check. Equations (43) and (44) then give

$$\begin{aligned}h_1' &= -\mathrm{i}\sigma h_0, \\ h_0' &= \frac{2h_0}{r} + \frac{\sigma^2 - 4/r^2}{\mathrm{i}\sigma}h_1.\end{aligned} \tag{45}$$

Eliminating h_0 and writing $h_1 = r\Psi$ yields

$$\Psi'' + \left(\sigma^2 - \frac{6}{r^2}\right)\Psi = 0, \tag{46}$$

the $J = 2$ Regge–Wheeler vacuum equation. This test fixed the relative signs before matter was introduced.

B. Both sidebands and the complete axial stress

At the common Fourier frequency σ, let u_+ multiply the perturbation of Φ and u_- the independent perturbation of Φ^*. With $c_V = 1/(2\sqrt{\pi})$, the direct scalar parts of the projected stress are

$$\tau_t = \frac{c_V F}{2}\left[(2\omega + \sigma)u_- - (2\omega - \sigma)u_+\right], \tag{47}$$

$$\tau_r = -\frac{\mathrm{i}c_V}{2}\left[F'(u_+ + u_-) - F(u_+' + u_-')\right], \tag{48}$$

$$\tau_2 = \mathrm{i}c_V F(u_+ + u_-). \tag{49}$$

The explicit metric variation in the stress tensor adds $-\mathcal{K}_0 h_0/2$ to the tA projection and $-\mathcal{K}_0 h_1/2$ to the rA projection, where for $\ell = 1$

$$\mathcal{K}_0 = \frac{1}{4\pi}\left[-\frac{\omega^2 F^2}{\alpha^2} + \frac{F'^2}{\gamma^2} + \frac{2F^2}{r^2} + \mu^2 F^2\right]. \tag{50}$$

There is no such term in the AB equation in Regge–Wheeler gauge because $h_2 = 0$. Equations (43), (44), and (48)–(49) are two first-order Einstein equations. Equation (42) is retained as a dependent constraint monitor rather than used to overdetermine the integration.

C. Six-state first-order system

Define

$$\boldsymbol{Y} = (h_0, h_1, u_+, v_+, u_-, v_-)^T, \qquad v_\pm = u'_\pm. \tag{51}$$

The tensor Einstein equation gives

$$h'_1 = -\mathrm{i}\sigma\frac{\gamma^2}{\alpha^2}h_0 + \left(\frac{\gamma'}{\gamma} - \frac{\alpha'}{\alpha}\right)h_1 + 8\pi\mathrm{i}c_V\gamma^2 F(u_+ + u_-). \tag{52}$$

For compactness, introduce

$$\mathcal{B} = \frac{\alpha''}{\alpha\gamma^2} - \frac{\alpha'\gamma'}{\alpha\gamma^3} - \frac{\gamma'}{r\gamma^3} + \frac{\alpha'}{r\alpha\gamma^2} + \frac{2}{r^2}, \quad \mathcal{D} = F'(u_+ + u_-) - F(v_+ + v_-). \tag{53}$$

The rA equation becomes

$$h'_0 = \frac{2h_0}{r} + \frac{1}{\mathrm{i}\sigma}\left[\sigma^2 h_1 - 2\alpha^2\mathcal{B}h_1 - 8\pi\alpha^2\mathcal{K}_0 h_1 - 8\pi\mathrm{i}c_V\alpha^2\mathcal{D}\right]. \tag{54}$$

Finally, define

$$\mathcal{R}_h = Fh'_1 + \left[2F' + F\left(\frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)\right]h_1, \qquad \kappa_V = -\mathrm{i}c_V. \tag{55}$$

The two Klein–Gordon equations are

$$v'_+ = -\left(\frac{2}{r} + \frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)v_+ - \gamma^2\left[\frac{(\omega-\sigma)^2}{\alpha^2} - \mu^2 - \frac{6}{r^2}\right]u_+ - \frac{\mathrm{i}\kappa_V\gamma^2 F(2\omega-\sigma)}{\alpha^2 r^2}h_0 + \frac{\kappa_V}{r^2}\mathcal{R}_h, \tag{56}$$

$$v'_- = -\left(\frac{2}{r} + \frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right)v_- - \gamma^2\left[\frac{(\omega+\sigma)^2}{\alpha^2} - \mu^2 - \frac{6}{r^2}\right]u_- + \frac{\mathrm{i}\kappa_V\gamma^2 F(2\omega+\sigma)}{\alpha^2 r^2}h_0 + \frac{\kappa_V}{r^2}\mathcal{R}_h. \tag{57}$$

Together with $u'_\pm = v_\pm$, Eqs. (52)–(57) form the promised local system $\boldsymbol{Y}' = \boldsymbol{A}(r,\sigma)\boldsymbol{Y}$. The apparent $1/\sigma$ in Eq. (54) reflects the chosen constraint reduction; a separate zero-frequency formulation is required for a stationary axial perturbation.

D. Center basis and channel sheets

Let $F = f_1 r + O(r^3)$ and $\alpha = \alpha_c + O(r^2)$. The three regular solutions have

$$h_0 = Hr^3 + O(r^5), \qquad h_1 = Br^4 + O(r^6), \qquad u_+ = Pr^2 + O(r^4), \qquad u_- = Nr^2 + O(r^4), \tag{58}$$

where

$$B = \frac{1}{4}\left[-\frac{\mathrm{i}\sigma}{\alpha_c^2}H + 8\pi \mathrm{i} c_V f_1 (P+N)\right]. \tag{59}$$

Thus (H, P, N) generate a three-column interior fundamental matrix. At infinity the gravitational channel is outgoing and the scalar wavenumbers are

$$k_\pm = \sqrt{(\omega \mp \sigma)^2 - \mu^2}, \tag{60}$$

with the sheet chosen so that $\operatorname{Im} k_\pm \geq 0$. An open channel is outgoing as $e^{+\mathrm{i}k_\pm r_*}$; a closed channel decays on the same sheet. For the low-frequency mode reported below both scalar channels are closed.

Finite-radius plane-wave conditions are inadequate for the very weakly damped gravitational solution. We therefore propagate h_0/h_1 and the two scalar logarithmic derivatives through the exact Schwarzschild exterior. The integration starts at $r = 300$ with the exact stable logarithmic derivative of the $J = 2$ spherical Hankel function. Linear channel pairs are evolved in short inward segments and renormalized after every segment; this avoids both Riccati poles and exponential overflow in a closed scalar channel.

E. Two-sided matching and numerical controls

Two-sided matching avoids the exponential contamination that affects direct outward construction of weakly damped or massive-field modes; this is the same boundary-value issue underlying established QNM integration and continued- fraction methods [4, 31].

Let Y_{in} be the 6×3 matrix obtained by integrating the three regular center solutions outward and Y_{out} the corresponding matrix obtained by integrating the three physical exterior channels inward. A mode exists when their column spaces intersect:

$$\det\left[Y_{\text{in}}(r_m), -Y_{\text{out}}(r_m)\right] = 0. \tag{61}$$

The smallest singular value of a Euclidean row- and column-scaled matrix is used only as a conditioning diagnostic in the coarse search. Because those norms contain complex conjugation, the scaled determinant is not holomorphic. Local Newton refinement uses the unnormalized determinant in Eq. (61). For root counting, each three-plane is evolved in $\Lambda^3(\mathbb{C}^6)$: its twenty 3×3 minors obey the induced compound-matrix equation, and the exterior product of the two evolved three-forms is a holomorphic Evans function. This prevents column collapse without introducing QR phases or Euclidean normalizations.

The production workflow searches $0.0488 \le \mathrm{Re}\,\sigma \le 0.0500$ and $-1.2\times10^{-6} \le \mathrm{Im}\,\sigma \le -10^{-7}$ on a 7×5 grid; no stored pole is read. On the smaller contour $0.04936 \le \mathrm{Re}\,\sigma \le 0.04944$ and $-1.1\times10^{-6} \le \mathrm{Im}\,\sigma \le -10^{-7}$, an eight-point-per-edge contour has winding one but a marginal maximum phase step of 2.8622 rad. Adaptive bisection of only under-resolved segments produces 38 boundary samples, preserves winding one, and reduces the maximum phase step to $1.55284 < \pi/2$. A stricter production rerun with a $\pi/4$ acceptance threshold uses 46 samples and gives a maximum step 0.60974, again with winding one. Expanding the local contour to $0.04930 \le \mathrm{Re}\,\sigma \le 0.04950$ and $-1.5\times10^{-6} \le \mathrm{Im}\,\sigma \le 10^{-7}$ gives winding one with 43 samples and maximum phase step $0.78087 < \pi/4$. These counts certify one zero only in the declared local region; they are not a complete spectrum or a global stability proof.

Five controls are evaluated before interpreting a root. First, $F = 0$ must give Eq. (46). Second, the two scalar equations must decouple to the massive flat-space $L = 2$ equations. Third, the propagated flat exterior ratios must reproduce the exact spherical-Hankel logarithmic derivatives. Fourth, the root must persist when both r_m and the radius at which the stellar integration enters the Schwarzschild exterior are moved. The first four controls pass. Fifth, the unused tA Einstein equation is evaluated on the mode using α'' reconstructed analytically from the equilibrium EKG equations. A refined equilibrium profile and the independent chain-rule evaluation $Y'' = A'Y + A^2Y$ give a stable relative L_2 defect of 2.4×10^{-7} and a pointwise maximum of 1.95×10^{-6}. Changing the derivative step by a factor of twenty leaves the largest residual near 2×10^{-6}. This is a two-order-of-magnitude improvement, but it is a resolved operator-level floor; we therefore do not mark the constraint gate closed.

F. First frequency-domain axial mode

For the validated $a_1^0 = 0.08$ background,

$$\omega = 0.8518974044, \qquad M = 1.1712711606, \qquad C_{99} = 0.1048865408. \tag{62}$$

TABLE III. Refined neutral $J = L = 2$ axial frequency-domain pole. The last column is the minimum singular value of the conditioned match matrix.

r_m	r_e	$\mathrm{Re}\,\sigma$	$\mathrm{Im}\,\sigma$	$s_{\min}$
14	35	0.0493977307853	$-5.96519340 \times 10^{-7}$	7.91×10^{-15}
12	30	0.0493977307847	$-5.96519314 \times 10^{-7}$	2.02×10^{-14}

Two independently refined matching setups give Table III. At fixed $r_{\rm far} = 600$, their mean and full spread are

$$\sigma = 0.0493977307850 - 5.96519327 \times 10^{-7}\mathrm{i}, \quad \Delta\sigma_R = 6.30 \times 10^{-13}, \quad \Delta\sigma_I = 2.65 \times 10^{-14}. \tag{63}$$

The corresponding quality factor is

$$Q = \frac{\sigma_R}{-2\sigma_I} = 4.1405 \times 10^4. \tag{64}$$

The exterior start now uses the Schwarzschild Regge–Wheeler master expansion and the massive-scalar Coulomb expansion through $O(r^{-3})$. For $r_{\rm far} = 120, 200, 300, 600, 900$, the imaginary parts are -5.9637944, -5.9650567, -5.9652382, -5.9651933, and -5.9651891, in units of 10^{-7}; the 600–900 shift is 4.24×10^{-13}. At $r_{\rm far} = 300$, increasing the asymptotic order from two to three shifts (σ_R, σ_I) by $(3.10, 4.74) \times 10^{-10}$. Treating that order shift conservatively, we quote

$$\sigma = 0.0493977307(4) - \mathrm{i}(5.9652 \pm 0.0048) \times 10^{-7}, \qquad Q \simeq 4.1405 \times 10^4, \tag{65}$$

where the displayed uncertainty is set by asymptotic order rather than the much smaller matching-domain spread. The first scalar threshold is $\sigma_{\rm th} = \mu - \omega = 0.1481025957$, well above the mode. Its scalar tails are therefore evanescent and the negative imaginary part represents gravitational leakage. This is the expected signature of a long-lived, matter-supported axial oscillation that is density-silent in the Newtonian projection but radiatively visible in general relativity.

Equation (63) is regenerated from the declared search and establishes a counted discrete pole of the released neutral operator. The remaining qualification is the resolved unused-Einstein residual and the absence of a genuinely independent damping calculation.

G. Seven-background stable-branch continuation

To test whether the pole is an isolated numerical feature, we continue it through seven independently solved stable backgrounds spanning $0.065 \le a_1^0 \le 0.095$ and $0.0953 \le C_{99} \le 0.1131$. At every point, two matching domains are refined separately and a local exterior-algebra contour is adaptively resolved with maximum phase increment below $\pi/3$. All seven contours have winding one, all modes remain below the first scalar threshold, and adjacent normalized six-state profiles have overlaps above 0.9991. Table IV and Fig. 1 therefore establish a continuously tracked mode branch rather than one isolated root.

TABLE IV. Counted stable-branch continuation. The damping column is displayed in units of 10^{-7}; O_- is the overlap with the preceding normalized six-state profile.

a_1^0	C_{99}	$\mathrm{Re}\,\sigma$	$-10^7\,\mathrm{Im}\,\sigma$	Q	O_-
0.065	0.095286	0.0432529917	3.72185	58107	—
0.070	0.098665	0.0453566970	4.41231	51400	0.999187
0.075	0.101859	0.0474032891	5.16048	45929	0.999272
0.080	0.104886	0.0493977307	5.96519	41405	0.999344
0.085	0.107760	0.0513442785	6.82511	37614	0.999406
0.090	0.110493	0.0532466187	7.73884	34402	0.999459
0.095	0.113098	0.0551079718	8.70499	31653	0.999505

The largest two-domain shifts over the campaign are 4.96×10^{-12} in the real part and 8.52×10^{-14} in the imaginary part. These are far below the single-background asymptotic-order systematic. The present interval does not reach the dilute Newtonian limit, so it establishes a relativistic stable branch but does not yet justify a weak-field damping exponent.

H. Time-domain real-frequency consistency check

We independently rewrite the same covariant perturbation equations as a first-order-in-time, second-order-in-space system for

$$(h_0, h_1, k, p, \pi, q, \chi), \qquad k = \partial_t h_1, \quad \pi = \partial_t p, \quad \chi = \partial_t q. \tag{66}$$

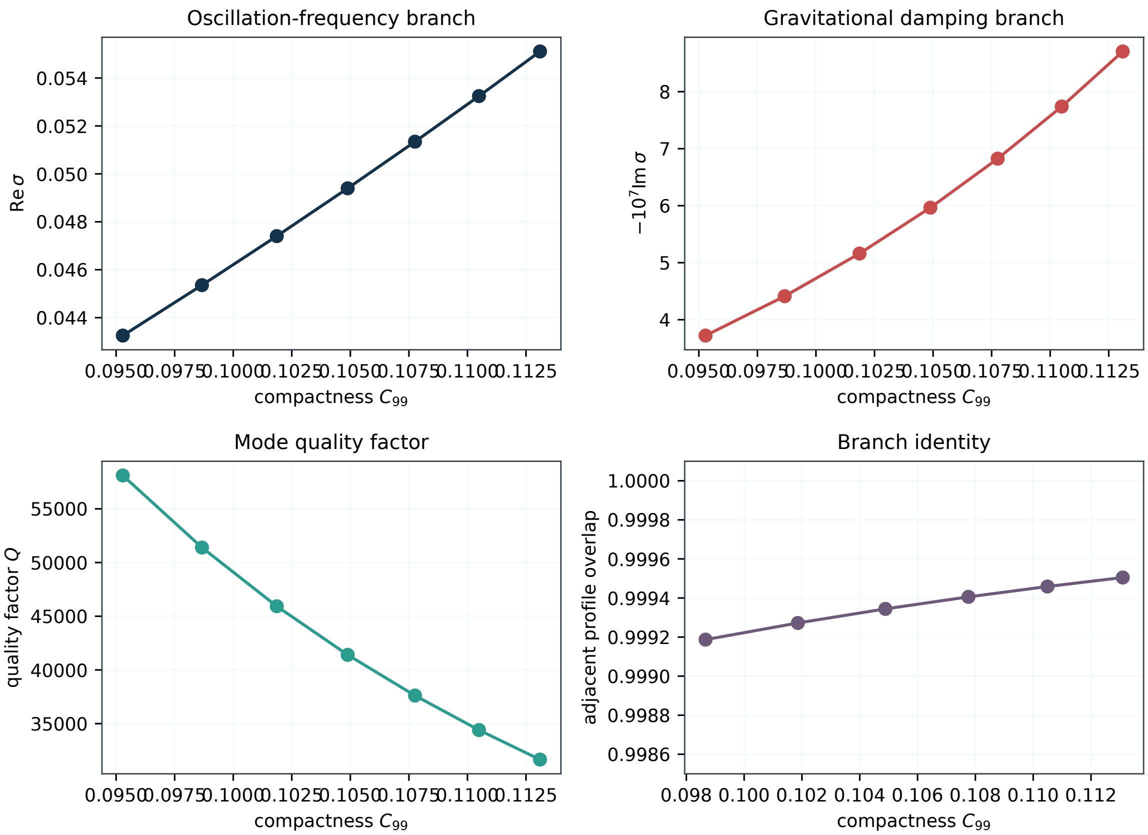

FIG. 1. Counted axial-mode branch across seven stable backgrounds. The real frequency and gravitational damping magnitude increase smoothly with compactness, while the quality factor decreases. Adjacent profile overlaps near unity independently preserve mode identity.

The AB Einstein equation determines $\partial_t h_0$, the rA equation evolves k, and the two Klein–Gordon equations evolve π and χ. Fourth-order finite differences, fourth-order Runge–Kutta integration, and a sixth-derivative Kreiss–Oliger filter are used. The stellar background is continued by exact Schwarzschild vacuum, and the outer boundary is placed at $r = 200$, more than one gravitational wavelength from the star. Data are fitted only before a boundary signal can return to the extraction radius.

As an operator identity, substitution of $e^{-\mathrm{i}\sigma t}$ into the time-domain equations reproduces Eqs. (52)–(57). With analytic radial derivatives the difference is at machine precision. The numerical evolution is initialized with the matched eigenfunction but does not evaluate the shooting determinant. Table V gives the resolution and Courant study. The regenerated time-domain matrix uses the current frequency-domain pole and the current evolution operator at every resolution and Courant factor. The four fitted frequencies span 4.74×10^{-8}; the maximum fractional error relative to the

TABLE V. Time-domain extraction at $r \simeq 6$ over $20 \le t \le 138$. All runs use $r_{\max} = 200$.

N_r	CFL	σ_R^{TD}	$\mathrm{d}\ln\lvert p\rvert/\mathrm{d}t$	phase RMS
1500	0.10	0.0493977844	-5.67×10^{-7}	2.371×10^{-6}
2000	0.10	0.0493977434	-5.98×10^{-7}	9.256×10^{-7}
2500	0.10	0.0493977370	-6.03×10^{-7}	7.897×10^{-7}
2000	0.08	0.0493977434	-5.98×10^{-7}	9.261×10^{-7}

frequency-domain value is 1.09×10^{-6} and decreases to 1.25×10^{-7} on the finest grid. The phase RMS lies between 7.90×10^{-7} and 2.37×10^{-6} rad. Because the initial data are the frequency-domain eigenfunction and the run length is only 9×10^{-5} of the $1/|\sigma_I| \simeq 1.68 \times 10^{6}$ damping time, the amplitude slopes cannot measure the leakage rate or certify its sign independently. This experiment checks only that the time-domain operator preserves the real oscillation frequency over the simulated window.

VI. SCATTERING, CONVERSION, AND DRIVEN RESPONSE

The complex root establishes the free oscillation. We now ask the reciprocal question: how does the same star respond when an axial wave is supplied from infinity? This calculation is more than a visualization of the QNM. It tests the incoming-wave phase convention and the interpretation of the narrow pole in a real-frequency experiment. The present open-channel weighting is an internally consistent scattering normalization, not yet an independent derivation of canonical gravitational and scalar energy flux.

A. Incoming channels and a flat-space unit-modulus identity

At real σ, let Y_g^{out} denote the gravitational exterior solution used in the QNM calculation. Complex conjugation alone does not produce its incoming partner in the first-order metric variables. With $e^{-\mathrm{i}\sigma t}$, the vacuum equation contains

$$h_1' = -\mathrm{i}\sigma \frac{\gamma^2}{\alpha^2} h_0 + \left(\frac{\gamma'}{\gamma} - \frac{\alpha'}{\alpha}\right) h_1. \tag{67}$$

Consequently radial reversal acts as

$$(h_0, h_1)_{\rm in} = (-h_0^*, h_1^*)_{\rm out}. \tag{68}$$

This sign is essential. In exact flat vacuum, matching a regular center to the in/out Hankel pair then gives $|S_{gg}| = 1$ to 2×10^{-7} in the regression control; using plain conjugation violates the vacuum Wronskian. For an incident scalar wave the exterior scalar pair is real-coefficient at fixed real frequency, so ordinary complex conjugation reverses its direction.

Below the first scalar threshold there is only one propagating channel. Over $0.02 \le \sigma \le 0.145$, including frequencies within a few widths of the long-lived pole, the physical solution obeys

$$\big||S_{gg}| - 1\big| < 4 \times 10^{-6}. \tag{69}$$

The closed scalar amplitudes can become large near the pole, but carry no asymptotic radial flux. Equation (69) is thus an independent global check of the center data, exterior propagation, and six-dimensional matching solve.

B. Raw two-channel scattering and inferred weighting

For $\sigma > \mu - \omega$, the u_- sideband opens and the exterior problem has two propagating channels. Endpoint-normalized amplitudes are collected as

$$\begin{pmatrix} b_g^{\rm out} \\ b_s^{\rm out} \end{pmatrix} = S \begin{pmatrix} b_g^{\rm in} \\ b_s^{\rm in} \end{pmatrix}, \qquad S = \begin{pmatrix} S_{gg} & S_{gs} \\ S_{sg} & S_{ss} \end{pmatrix}. \tag{70}$$

The raw basis uses $h_1(r_e) = 1$ for gravity and $u_-(r_e) = 1$ for the scalar; these columns do not carry equal flux. We write a positive diagonal weight $W = \mathrm{diag}(w_g, w_s)$. Reciprocity of the computed matrix supplies the ratio

$$\frac{w_s}{w_g} = -\frac{S_{gg}^* S_{gs}}{S_{sg}^* S_{ss}}, \tag{71}$$

and the imaginary fraction of the right-hand side is below 10^{-6} in every production run. The two diagonal equations in $S^\dagger W S = W$ are not used to determine the ratio and provide algebraic consistency tests. However, Eq. (71) infers W from S; it does not derive energy currents from the quadratic action, covariant symplectic current, or asymptotic Isaacson and Klein–Gordon fluxes. We therefore call the following quantities weighted conversion diagnostics, not canonical efficiencies.

The weighted conversion diagnostics are

$$\mathcal{P}_{g\to s} = \frac{w_s}{w_g}|S_{sg}|^2, \qquad \mathcal{P}_{g\to g} = |S_{gg}|^2,$$
$$\mathcal{P}_{s\to g} = \frac{w_g}{w_s}|S_{gs}|^2, \qquad \mathcal{P}_{s\to s} = |S_{ss}|^2. \tag{72}$$

Table VI shows that conversion is small but nonzero, reciprocal, and largest near $\sigma \simeq 0.20$ for the benchmark scan.

TABLE VI. Reciprocity-weighted axial conversion above the first scalar threshold. The last column is the spectral norm of $\widetilde{S}^\dagger\widetilde{S} - I$, where $\widetilde{S} = W^{1/2}SW^{-1/2}$.

σ	$\mathcal{P}_{g\to s}$	$\mathcal{P}_{s\to g}$	$\mathcal{P}_{g\to g} + \mathcal{P}_{g\to s}$	inferred-W isometry defect
0.16	3.7215×10^{-6}	3.7215×10^{-6}	1.000000022	2.21×10^{-8}
0.18	7.4777×10^{-6}	7.4777×10^{-6}	0.999999993	7.10×10^{-9}
0.20	1.02057×10^{-5}	1.02057×10^{-5}	1.000000009	8.59×10^{-9}
0.25	9.9322×10^{-6}	9.9322×10^{-6}	0.999999956	4.44×10^{-8}
0.30	5.7625×10^{-6}	5.7625×10^{-6}	0.999999976	2.44×10^{-8}

At $\sigma = 0.20$, the three domain pairs $(r_m, r_e) = (12, 30), (14, 35)$, and $(16, 40)$ give $\mathcal{P}_{g\to s} = 1.020999 \times 10^{-5}$, 1.020572×10^{-5}, and 1.020566×10^{-5}. The corresponding linear matching residuals range from 7.6×10^{-12} to 5.2×10^{-10}. This domain audit separates the physical five-decimal weighted conversion diagnostic from the much smaller algebraic balance defect.

C. The QNM as a driven resonance

We next drive the system with a unit incident gravitational wave on a fixed 41-point grid, $0.0493945 \le \sigma \le 0.0494010$, evaluated without reading the stored complex pole. For each solution we integrate the scalar storage functional

$$\mathcal{N}_s(\sigma) = \int_0^{r_m} \alpha\gamma r^2 \left(|u_+|^2 + |u_-|^2\right) \mathrm{d}r. \tag{73}$$

A Lorentzian plus a quadratic local floor fits the numerical response with normalized RMS residual 9.38×10^{-6}. Repeating the fit with linear and quadratic backgrounds and three edge trims supplies the fit-window audit. The primary center and half-width are

$$\sigma_c^{\mathrm{drive}} = 0.0493977308380, \qquad \Gamma_{1/2}^{\mathrm{drive}} = 5.9649253 \times 10^{-7}. \tag{74}$$

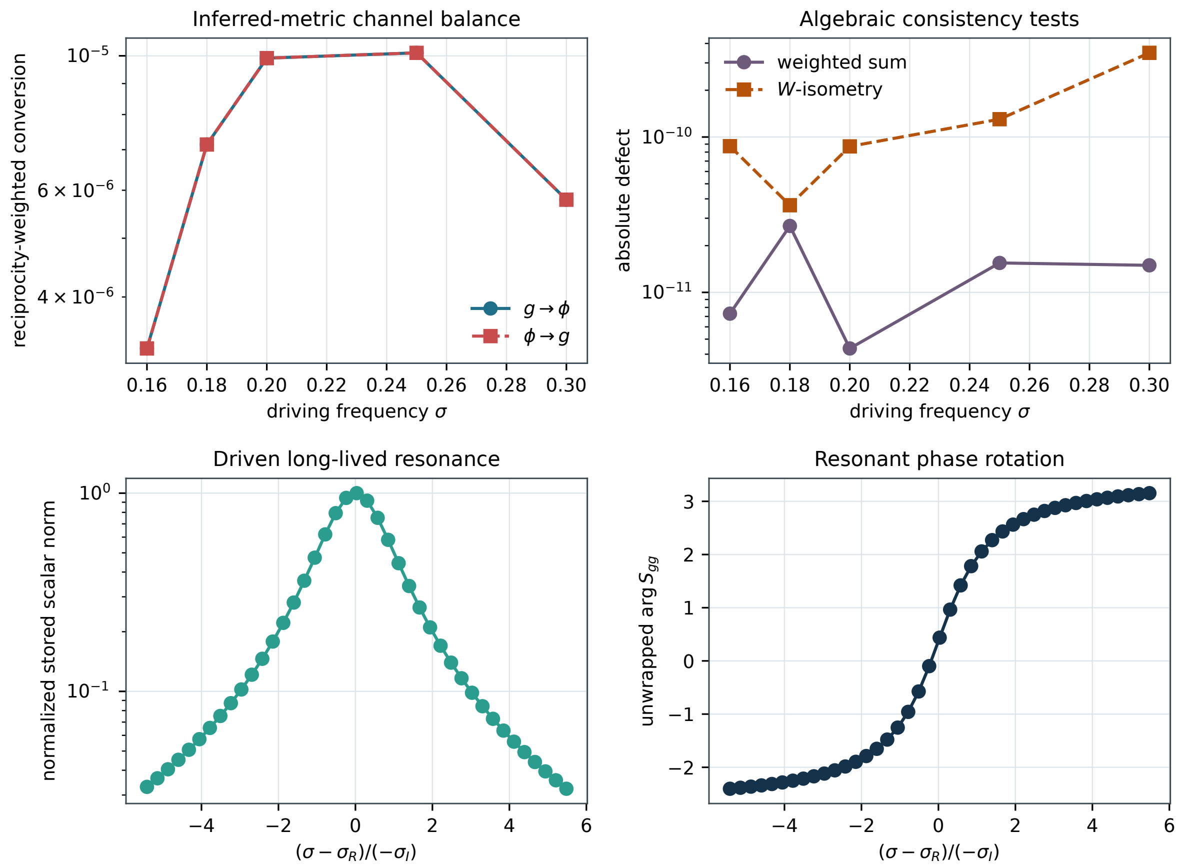

FIG. 2. Neutral axial response. Top left: conversion balanced by the positive diagonal reciprocity metric inferred from the numerical scattering matrix. Top right: weighted-sum and inferred-metric isometry defects; these are algebraic checks, not a canonical energy-flux derivation. Bottom left: stored scalar norm in a unit-incident-gravity experiment across the long-lived pole. Bottom right: the unwrapped reflection phase, whose rapid rotation gives the resonant delay.

with covariance errors 3.7×10^{-12} and 9.9×10^{-12}, respectively. The driven center differs from $\mathrm{Re}\, \sigma_{\mathrm{QNM}}$ by 1.1×10^{-9} fractionally, while the half-width differs from $-\mathrm{Im}\, \sigma_{\mathrm{QNM}}$ by 4.5×10^{-5}. The reflection-phase slope agrees with the single-pole prediction $2/\Gamma_{1/2}$ to 2.46%. This is strong same-operator evidence for the long-lived pole; the short time evolution is not counted as a damping measurement.

VII. STATIONARY AXIAL RESPONSE

The zero-frequency odd-parity sector is the relativistic gravitomagnetic response channel. Its interpretation follows the magnetic-type response defined for spherical compact bodies [32, 33]; boson-star tidal observables are independently known to retain information that distinguishes horizonless scalar configurations from black holes and material stars [34].

The constraint-reduced dynamical equations contain $1/\sigma$ and cannot be continued by numerical substitution to $\sigma = 0$. We instead return to the tA and AB Einstein equations before division by frequency. The regular stationary sector has

$$h_1 = 0, \qquad u_+ + u_- = 0, \qquad d = u_+ - u_-, \tag{75}$$

and closes on (h_0, h_0', d, d'). The scalar difference satisfies

$$d'' + \left(\frac{2}{r} + \frac{\alpha'}{\alpha} - \frac{\gamma'}{\gamma}\right) d' + \gamma^2 \left(\frac{\omega^2}{\alpha^2} - \mu^2 - \frac{6}{r^2}\right) d = -\frac{4c_V\omega\gamma^2 F}{\alpha^2 r^2} h_0, \tag{76}$$

while the tA equation supplies the second-order metric equation. At the center the two regular solutions begin as $h_0 \sim r^3$ and $d \sim r^2$.

In the Schwarzschild exterior the metric equation is

$$h_0'' - \frac{6r - 4M}{r^2(r - 2M)} h_0 = 0. \tag{77}$$

We use the exact growing solution $g = r^2(r - 2M)$ and the reduction-of-order decaying solution normalized as $d_g = r^{-2}[1 + O(M/r)]$. The regular interior two-plane and the exterior tidal, response, and decaying-scalar three-plane are matched at an independently varied radius. The metric at the outer normalization radius is decomposed as

$$h_0 = A\, r^2(r - 2M) + B\, r^{-2}[1 + O(M/r)]. \tag{78}$$

Our fixed-normalization response coefficient is the explicitly defined ratio

$$\Lambda_B = \frac{B}{AM^5} = -122.47418, \tag{79}$$

Varying $r_m = 10, 12, 14, 16$ at fixed outer normalization changes the coefficient by 8.4×10^{-7}; the complete six-domain spread, including outer radii $30, 35, 40$, is 3.15×10^{-5}, or 2.6×10^{-7} fractionally. The two-sided matching residual is 3×10^{-16}–3×10^{-15} and the coefficient is real to machine precision. We call Λ_B an explicitly normalized axial response coefficient rather than a gauge-invariant magnetic Love number; a separate gauge-invariant reconstruction is required before making that identification.

VIII. RELATIVISTIC RADIAL PERTURBATIONS

A. Symmetry, harmonic dependence, and spectral character

Radial perturbations preserve the diagonal spherical symmetry of the background. They do not carry the tensor radiative degrees of freedom that appear in nonspherical sectors, so the present problem is a normal-mode stability problem rather than an outgoing gravitational-wave QNM problem. All perturbed amplitudes are taken proportional to $\exp(-\mathrm{i}\sigma t)$. Time-reversal invariance and the real background equations organize the reduced system in the parameter σ^2.

For $\sigma^2 > 0$, the pair $\pm\sigma$ describes the same oscillatory radial mode with opposite time phase. At $\sigma^2 = 0$, the mode is marginal and identifies the exchange of radial stability. For $\sigma^2 < 0$, writing $\sigma = \mathrm{i}\Gamma$ gives exponentially growing and decaying solutions with $\Gamma = \sqrt{-\sigma^2}$. This is why the stability plots retain signed σ^2 rather than taking a square root prematurely.

The radial coordinate and time normalization are inherited from the equilibrium ansatz. A rescaling of the asymptotic lapse would rescale both ω and σ; enforcing the Schwarzschild lapse at the outer boundary fixes that freedom. The remaining eigenfunction normalization is arbitrary because the perturbation equations are homogeneous. We fix only a center amplitude for the nonlinear BVP and a maximum component for the spectral eigenvector, then remove this convention before comparing physical-field shapes.

The use of a finite outer boundary does not impose an outgoing-wave condition. The radial modes considered here are bound pulsations of a localized massive configuration. The published physical conditions constrain the scalar and metric perturbations consistently with the finite-domain equilibrium tail. A complex-frequency nonradial problem will require a qualitatively different boundary analysis, including open gravitational channels and propagating or evanescent scalar sidebands.

B. Variables and center behavior

Following Ref. [2], we evolve the numerical variables $\delta\varphi_{11}$ and

$$\delta L = \frac{\delta\lambda}{2\kappa\psi_1^2}, \tag{80}$$

where $\delta\lambda$ is the physical radial metric perturbation. The first-order state is

$$\boldsymbol{y} = (\delta\varphi, \delta\varphi', \delta L, \delta L')^T, \qquad \boldsymbol{y}' = \boldsymbol{F}(r, \boldsymbol{y}; \sigma^2). \tag{81}$$

To display the implemented operator compactly, define

$$q = \frac{\psi'}{\psi}, \qquad a = \frac{\alpha'}{\alpha}, \qquad g = \frac{\gamma'}{\gamma}, \qquad \mu_\ell^2 = 1 + \frac{\ell(\ell+1)}{r^2}. \tag{82}$$

The scalar perturbation equation is

$$\delta\varphi'' = -\left(\frac{2}{r} + a - g\right)\delta\varphi' - \frac{2}{r}\delta L' + \mathcal{P}_\varphi \delta\varphi + \mathcal{P}_L \delta L, \tag{83}$$

$$\mathcal{P}_\varphi = 2\gamma^2 \left[\frac{q^2}{\gamma^2} + \kappa r \mu_\ell^2 \psi\psi' + \mu_\ell^2 + \frac{2\omega^2 - \sigma^2}{2\alpha^2}\right], \tag{84}$$

$$\mathcal{P}_L = -2\left[\frac{1}{r^2} + \frac{2}{r}(q-g) + \kappa\psi\psi'\left(a - g + q + \frac{1}{r}\right) - \kappa\gamma^2\psi^2\left(\mu_\ell^2 - \frac{\omega^2}{\alpha^2}\right)\right]. \tag{85}$$

The metric-sector equation is

$$\delta L'' = \mathcal{Q}_{\varphi'}\delta\varphi' + \mathcal{Q}_{L'}\delta L' + \mathcal{Q}_\varphi \delta\varphi + \mathcal{Q}_L \delta L, \tag{86}$$

$$\mathcal{Q}_{\varphi'} = -2\left(2q - r\gamma^2\mu_\ell^2\right), \qquad \mathcal{Q}_{L'} = -\left[4q + 3(a-g)\right], \tag{87}$$

$$\mathcal{Q}_\varphi = -2\gamma^2\left[\frac{2q^2}{\gamma^2} - r\mu_\ell^2(2q + 2a + g) + \frac{\ell(\ell+1)}{r^2}\right], \tag{88}$$

$$\begin{aligned}\mathcal{Q}_L = 2\Bigg[&2\kappa\psi'^2 - \left(q - \frac{1}{r} + a - g\right)^2 + \frac{2}{r^2} - \frac{4a - g}{r} + g' \\ &-\gamma^2\left(\mu_\ell^2 - \frac{2\omega^2 - \sigma^2}{2\alpha^2}\right)\Bigg].\end{aligned} \tag{89}$$

All background derivatives in these expressions are reconstructed through Eqs. (9)–(12). In particular, ψ'' and g' are evaluated from differential identities rather than from finite differences of stored profiles. The apparent powers of $1/r$ and q cancel when the regular background and perturbation series are substituted.

C. Frobenius data and normalization

Near the center, the arbitrary overall mode normalization is fixed by

$$\delta\varphi = 1 + cr^2 + O(r^4), \qquad \delta\varphi' = 2cr + O(r^3), \tag{90}$$

$$\delta L = 1 + dr^2 + O(r^4), \qquad \delta L' = 2dr + O(r^3), \tag{91}$$

with

$$d = \frac{1}{5}\left[1 + \frac{2\omega^2 - \sigma^2}{2\alpha_c^2} + 6\kappa\left(\frac{a_1^0}{\kappa}\right)^2\right]. \tag{92}$$

The coefficient in Eq. (92) must use the actual leading coefficient of the radial field in the implemented normalization. Replacing a_1^0/κ by an unconverted tabulated amplitude leaves a nonvanishing center-operator defect.

The coefficient c is not fixed locally. Together with σ^2, it is determined by the two outer conditions. For the nonlinear BVP we normalize both leading values to unity and impose the center relations at $r = \epsilon$. For the generalized eigenproblem, the arbitrary normalization must remain homogeneous. Defining

$$\mathcal{L}_\epsilon[f] = f(\epsilon) - \frac{\epsilon}{2} f'(\epsilon), \tag{93}$$

the center rows become

$$\mathcal{L}_\epsilon[\delta L] - \mathcal{L}_\epsilon[\delta\varphi] = 0, \tag{94}$$

$$\delta L'(\epsilon) - 2\epsilon d_0 \mathcal{L}_\epsilon[\delta\varphi] = -\sigma^2 \frac{\epsilon}{5\alpha_c^2} \mathcal{L}_\epsilon[\delta\varphi], \tag{95}$$

where $d_0 = d(\sigma^2 = 0)$. This form places the eigenvalue dependence explicitly in the right-hand matrix of the pencil.

D. Physical outer conditions

At a finite outer radius, the published physical conditions can be written

$$F_1 = \sqrt{\frac{\gamma}{\alpha}}\, r\psi\, \delta\varphi = 0, \tag{96}$$

$$F_2 = \sqrt{\frac{\alpha}{\gamma}}\, \frac{\psi \delta L - r\psi' \delta\varphi}{\omega} = 0. \tag{97}$$

Because ψ and ω are nonzero at the finite boundary, these conditions are equivalent to Dirichlet data for both numerical variables in the spectral problem. The nonlinear boundary-value formulation imposes $F_1 = F_2 = 0$ directly. Endpoint values are reported, but because they are imposed equations they are not interpreted as independent error estimates.

E. Physical reconstruction and nodes

The two fields used for cross-method comparison are

$$\delta\Phi_{\text{phys}} = \psi\, \delta\varphi, \qquad \delta\lambda = 2\kappa\psi^2 \delta L. \tag{98}$$

Interior zeros of $\delta\lambda$ are located by shape-preserving interpolation and bracketing. Only samples at the level of floating-point and interpolation noise are removed; the former fixed threshold $10^{-4}\max|\delta\lambda|$ is not used because it merges low-amplitude lobes of higher modes.

IX. NUMERICAL METHODS

A. Global nonlinear boundary-value solve

The first discretization treats Eq. (81) as a nonlinear global boundary-value problem with σ^2 as a free parameter. The center series supplies three independent regularity conditions after fixing the normalization, and the two physical outer equations complete the five conditions required for four first-order fields plus one eigenvalue. An outward initial-value integration is used only to seed Newton iteration; it is not used to certify the outer boundary.

At the inner boundary $r = \epsilon$, the three conditions are

$$\delta\varphi - \frac{\epsilon}{2}\delta\varphi' = 1, \tag{99}$$

$$\delta L - \frac{\epsilon}{2}\delta L' = 1, \tag{100}$$

$$\delta L' - 2\epsilon d(\sigma^2) = 0. \tag{101}$$

Together with $F_1 = F_2 = 0$ at $r = r_{\max}$, these form a five-component boundary vector for the four fields and the unknown parameter σ^2. The mode normalization is arbitrary; choosing unit leading values makes the initial Newton iterate and the cross-method reconstruction reproducible without changing the eigenvalue.

The solver returns an adaptive collocation polynomial. We record the maximum interval RMS relative residual reported by `solve_bvp`, and separately sample

$$\mathcal{R}_\infty = \max_{r,i} \frac{|y'_{i,\mathrm{coll}}(r) - F_i(r, \boldsymbol{y}; \sigma^2)|}{1 + |F_i(r, \boldsymbol{y}; \sigma^2)|} \tag{102}$$

on a 3000-point geometric grid. The terminology is explicit because an interval RMS residual is not a maximum pointwise differential-equation defect.

The earlier affine outward-basis method is deprecated. It formed linear combinations of exponentially amplified solutions and could reproduce an interior eigenfunction while generating misleading normalized boundary residuals through catastrophic cancellation. No number in the present certification uses that formulation.

B. Independent Chebyshev generalized eigenproblem

The second discretization uses N Chebyshev–Lobatto points on $[\epsilon, r_{\max}]$ [35, 36],

$$x_j = \cos\frac{\pi j}{N-1}, \qquad r_j = \frac{r_{\max}+\epsilon}{2} + \frac{r_{\max}-\epsilon}{2} x_j, \tag{103}$$

with first- and second-derivative matrices D and D^2. The coupled second-order equations are linear in σ^2, producing

$$A\boldsymbol{x} = \sigma^2 B\boldsymbol{x}. \tag{104}$$

The unknown vector is

$$\boldsymbol{x} = (\delta\varphi_0, \ldots, \delta\varphi_{N-1}, \delta L_0, \ldots, \delta L_{N-1})^T. \tag{105}$$

For $i \neq j$, the unmapped Chebyshev differentiation matrix is constructed from

$$D^{(x)}_{ij} = \frac{c_i}{c_j}\frac{(-1)^{i+j}}{x_i - x_j}, \qquad c_0 = c_{N-1} = 2, \quad c_j = 1, \tag{106}$$

with diagonal entries fixed by the zero row-sum property and the affine factor $2/(r_{\max}-\epsilon)$. The two differential equations fill $2(N-2)$ interior rows. The homogeneous center relations (94)–(95) and two Dirichlet outer rows complete the $2N \times 2N$ pencil. The spectral coefficient blocks and the $\sigma^2 = 0$ center coefficient are transcribed in a dedicated module that does not call the boundary-value right-hand side or center function. Tests evaluate the assembled matrix on arbitrary smooth fields and compare every interior, center, and outer row with a direct differential-operator calculation.

One dense Chebyshev domain becomes ill-scaled as N increases. Before solving Eq. (104), positive diagonal row and column scalings are applied,

$$A_s = D_r A D_c, \qquad B_s = D_r B D_c, \qquad A_s\boldsymbol{v} = \sigma^2 B_s \boldsymbol{v}. \tag{107}$$

The right eigenvector is transformed back with $\boldsymbol{x} = D_c\boldsymbol{v}$. We report both scaled and unscaled residuals,

$$\rho(A, B; \boldsymbol{x}) = \frac{\|A\boldsymbol{x} - \sigma^2 B\boldsymbol{x}\|_2}{\|A\boldsymbol{x}\|_2 + |\sigma^2|\,\|B\boldsymbol{x}\|_2}, \tag{108}$$

and a left/right condition diagnostic for the equilibrated pencil,

$$\kappa_\lambda = \frac{\|\boldsymbol{w}\|_2\|\boldsymbol{v}\|_2}{|\boldsymbol{w}^\dagger B_s \boldsymbol{v}|}. \tag{109}$$

Small algebraic residual establishes that the finite pencil was solved accurately; it does not by itself bound continuum error.

The finite pencil is singular in B because the eigenvalue enters only selected field and center rows. Generalized eigensolution therefore returns finite physical candidates together with infinite algebraic eigenvalues. We discard nonfinite values, require imaginary contamination below 10^{-7}, retain a prescribed real interval, reconstruct the physical fields, and sort the finite modes by σ^2. This filtering is deliberately simple: mode identity is decided by continuity, nodes, and overlap rather than by the raw ordering of a dense eigensolver.

C. Background smoothness, filtering, and continuation

The official spectral solver requires a C^1 cubic-Hermite reconstruction of the regular field $u = \psi/r^\ell$ from the stored ψ and ψ'. PCHIP is rejected by the public spectral API. Although PCHIP can yield a small matrix residual, its piecewise-smooth derivatives bias the continuum eigenvalues, especially for overtones. It remains available only as a local boundary-value representation variation.

Finite eigenvalues are retained if their imaginary contamination is below tolerance and their real part lies inside the requested interval. Mode identity combines frequency continuity, resolved nodal structure, and the normalized physical overlap

$$O[f, g] = \frac{|\int f(r)g(r)\mathrm{d}r|}{\left(\int f^2\mathrm{d}r\right)^{1/2}\left(\int g^2\mathrm{d}r\right)^{1/2}}. \tag{110}$$

The arbitrary real sign of an eigenvector is fixed by the sign of the overlap before plotting.

D. Uncertainty policy

The ground-mode budget contains the $r_{\max} = 30$ to 40 boundary-value shift, the boundary-value to $N = 80$ spectral difference, the maximum displacement of the $N = 50$–160 spectral sequence from the boundary-value reference, and the Hermite-to-PCHIP local boundary-value shift. The overtone budget is separate because its outer-domain dependence is stronger. Since these are deterministic systematic variations with no demonstrated probabilistic independence, the conservative sum is the quoted uncertainty.

E. End-to-end computational workflow

The production calculation is organized as a sequence of transformations whose intermediate outputs can each be checked. For an equilibrium amplitude a_1^0, the workflow is:

1. solve the nonlinear equilibrium boundary-value problem for (M, α, ψ, ψ') and ω;
2. evaluate the solution on a dense geometric grid and compute mass fractions, Noether charge, compactness, residuals, and stress-energy diagnostics;
3. construct a regular C^1 representation of $u = \psi/r^\ell$ and the even metric profiles;
4. solve the radial mode as a global nonlinear BVP, using outward integration only to seed the Newton iteration;
5. assemble the radial coefficient matrix independently on a Chebyshev grid, equilibrate the generalized pencil, and compute left and right eigenvectors;
6. filter finite nearly real eigenvalues, reconstruct physical fields, count nodes, and continue modes by overlap;
7. compare common-domain eigenvalues and eigenfunctions before combining controlled numerical variations into deterministic bounds.

No later stage overwrites an earlier diagnostic. In particular, a successful eigenvalue solve does not erase a failed background residual, and a small algebraic pencil residual does not replace domain or resolution convergence.

The equilibrium continuation proceeds outward from the validated $a_1^0 = 0.08$ anchor in separate low- and high-amplitude directions. At each step, the preceding converged profile is interpolated to the new adaptive mesh and the scalar amplitude is rescaled to the requested central value. This prevents the nonlinear solver from jumping to a radially excited branch or to a different segment of the relativistic spiral. The frequency is represented by an unconstrained logistic parameter, keeping $0 < \omega < 1$ during Newton updates rather than clipping an unconstrained eigenfrequency after the fact.

For the radial BVP, the unknown σ^2 is a solver parameter. The three center relations and two physical endpoint conditions supply five boundary equations for four fields plus that parameter. This global formulation is important because the two independent solutions of the coupled second-order system can acquire exponentially large components under outward integration. A shooting residual may then be a small difference of large numbers, whereas collocation enforces both endpoints as part of one nonlinear system.

The spectral calculation has a different numerical failure mode. With two fields on N Lobatto points, A and B are dense $2N \times 2N$ matrices. Storage grows as $O(N^2)$ and the full generalized eigensolve as $O(N^3)$. Increasing N therefore raises cost and can amplify coefficient and roundoff sensitivity even while the interpolation error decreases. Row and column equilibration reduce scale disparities but cannot change the intrinsic conditioning of a physical eigenpair. This is why the condition diagnostic and the unscaled residual are retained together.

F. Verification hierarchy

The tests are arranged in a hierarchy from local algebra to global physics:

1. analytic center-series coefficients and parity;
2. Chebyshev differentiation of known polynomials;
3. equality of independently assembled coefficient blocks at sampled radii;
4. consistency between differentiated background interpolants and the equilibrium equations;
5. endpoint conditions, residual definitions, and node counting;
6. domain, mesh, tolerance, cutoff, and spectral-resolution variations;
7. cross-method eigenvalues and physical-field overlaps;
8. literature frequencies, turning-point sign change, and overtone morphology.

A failure at a low level invalidates the higher-level comparison. Conversely, a passed literature benchmark cannot by itself reveal a local transcription error that happens to cancel at one model. The hierarchy is designed so that algebraic, numerical, and physical checks constrain different failure modes.

X. RESULTS

A. Equilibrium sequence

Figure 3 shows the 21-model $\ell = 1$ sequence. The grid maximum occurs at $a_1^0 = 0.10$, with $\omega = 0.8356722$ and $M_T = 1.1763866$, consistent with the published values $\omega \simeq 0.836$ and

$M \simeq 1.18$ [2]. The $a_1^0 = 0.08$ benchmark gives $\omega = 0.8518974$, reproducing the tabulated 0.8519.

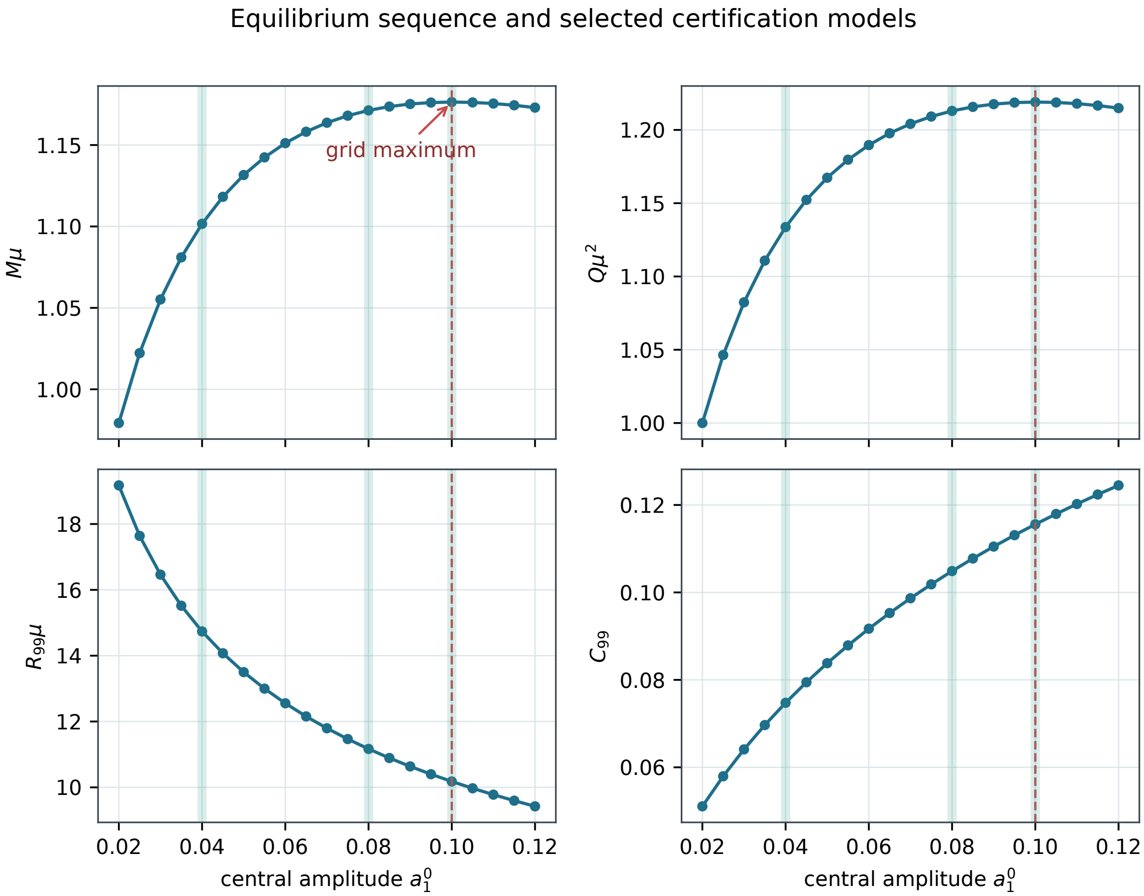

FIG. 3. Global $\ell = 1$ equilibrium sequence. The four panels show ADM mass, Noether charge, R_{99}, and C_{99} versus central amplitude. The dashed line marks the maximum-mass grid point; pale vertical bands identify representative profiles used below.

TABLE VII. Representative equilibrium models.

Model	a_1^0	ω	M_T	Q	R_{99}	C_{99}
Dilute	0.040	0.8964258	1.1016307	1.1337079	14.7365	0.07476
Benchmark	0.080	0.8518974	1.1712712	1.2129650	11.1673	0.10489
Maximum grid	0.100	0.8356722	1.1763866	1.2190106	10.1782	0.11558

Table VIII reports the complete production sequence rather than only the three configurations highlighted in the figures. The monotonic decrease of ω and R_{99} is accompanied by a mass maximum between the $a_1^0 = 0.10$ and 0.105 samples. The Noether charge reaches its maximum at the same grid point to the precision resolved here.

TABLE VIII: Complete 21-model $\ell = 1$ equilibrium sequence.

a_1^0	ω	M_T	Q	R_{99}	R_{999}	C_{99}
0.020	0.9302456	0.9793151	1.0000186	19.1736	23.5241	0.05108
0.025	0.9205267	1.0222457	1.0464047	17.6390	21.6681	0.05795
0.030	0.9117745	1.0551731	1.0823419	16.4625	20.2471	0.06410
0.035	0.9037867	1.0810265	1.1108190	15.5184	19.1071	0.06966
0.040	0.8964258	1.1016307	1.1337079	14.7365	18.1648	0.07476
0.045	0.8895929	1.1182024	1.1522636	14.0734	17.3666	0.07946
0.050	0.8832135	1.1315939	1.1673702	13.5009	16.6774	0.08382
0.055	0.8772299	1.1424256	1.1796748	12.9988	16.0739	0.08789
0.060	0.8715961	1.1511625	1.1896659	12.5539	15.5395	0.09170
0.065	0.8662747	1.1581622	1.1977207	12.1551	15.0605	0.09528
0.070	0.8612345	1.1637050	1.2041372	11.7946	14.6296	0.09866
0.075	0.8564494	1.1680143	1.2091544	11.4672	14.2372	0.10186
0.080	0.8518974	1.1712712	1.2129669	11.1673	13.8784	0.10488
0.085	0.8475592	1.1736237	1.2157352	10.8911	13.5477	0.10776
0.090	0.8434184	1.1751947	1.2175930	10.6363	13.2436	0.11049
0.095	0.8394603	1.1760869	1.2186530	10.3989	12.9606	0.11310
0.100	0.8356722	1.1763866	1.2190106	10.1782	12.6970	0.11558
0.105	0.8320426	1.1761676	1.2187478	9.9716	12.4510	0.11795
0.110	0.8285614	1.1754927	1.2179347	9.7777	12.2204	0.12022
0.115	0.8252196	1.1744160	1.2166324	9.5951	12.0034	0.12240
0.120	0.8220088	1.1729844	1.2148941	9.4232	11.7987	0.12448

Representative profiles are shown in Fig. 4. Increasing central amplitude narrows the scalar distribution, deepens the central lapse, increases the radial metric peak, and moves the enclosed mass inward. The trend is smooth along the sampled branch and does not show resolved radial nodes.

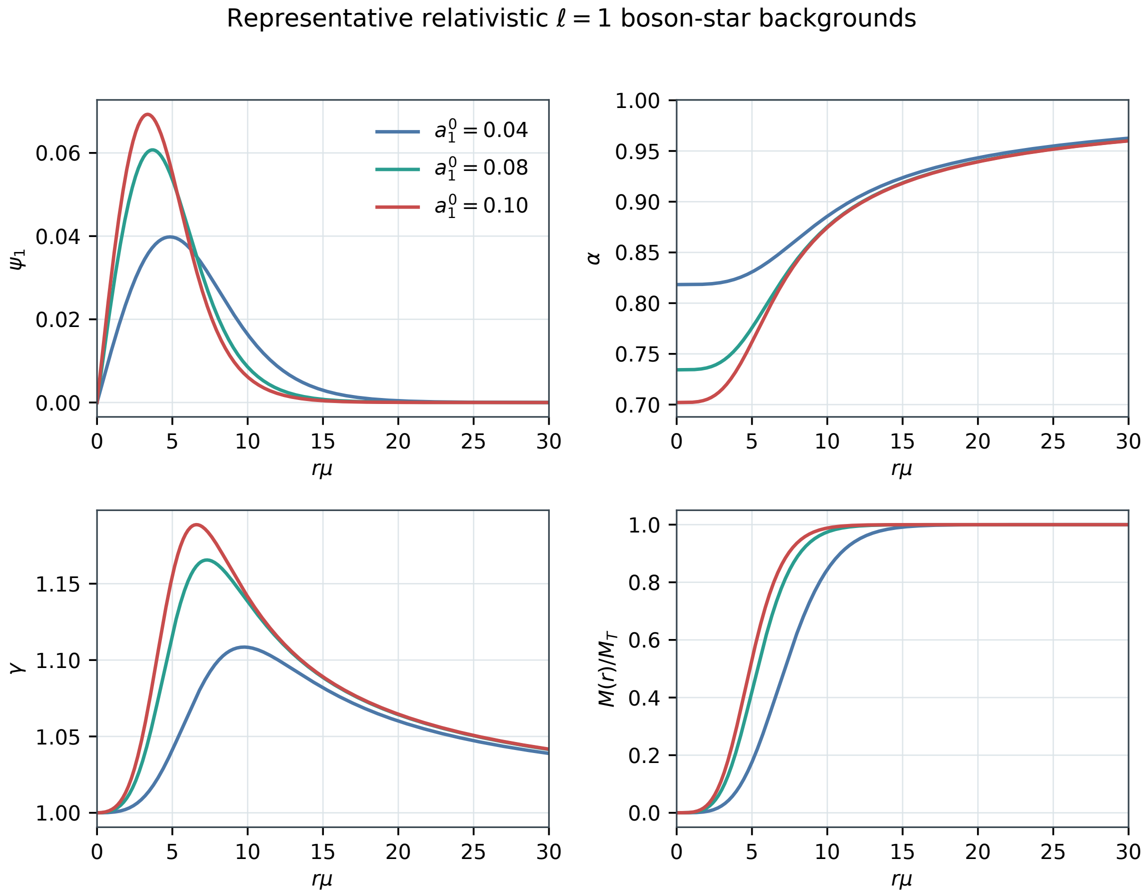

FIG. 4. Scalar field, lapse, radial metric function, and enclosed mass for dilute, benchmark, and maximum-mass-grid configurations.

Figure 5 separates two notions that are sometimes conflated. The entire sampled branch is energetically bound according to $M_T - Q < 0$, whereas the dynamical ground mode changes sign only at the maximum-mass turning point. The right panel overlays a finite-difference estimate of dM_T/da_1^0 with the independently computed σ_0^2 sequence.

The dense background residual and normalized outer-tail residual are displayed in Fig. 6. The first rises gradually as the configurations become more compact but remains below 2×10^{-5} throughout the scan. The Robin residual is generally near floating-point precision after normalization; the isolated larger value at the benchmark is still negligible compared with the profile amplitude and has no visible effect on the outer-domain comparison.

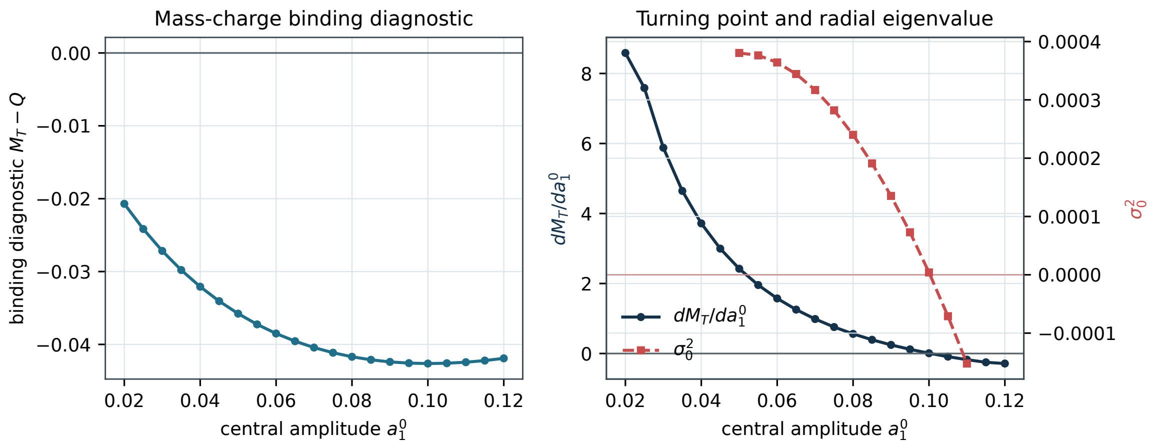

FIG. 5. Energetic and dynamical diagnostics. Left: mass-charge binding diagnostic along the full sequence. Right: finite-difference mass derivative and the radial ground eigenvalue. Their zero crossings agree at the resolution of the amplitude grid.

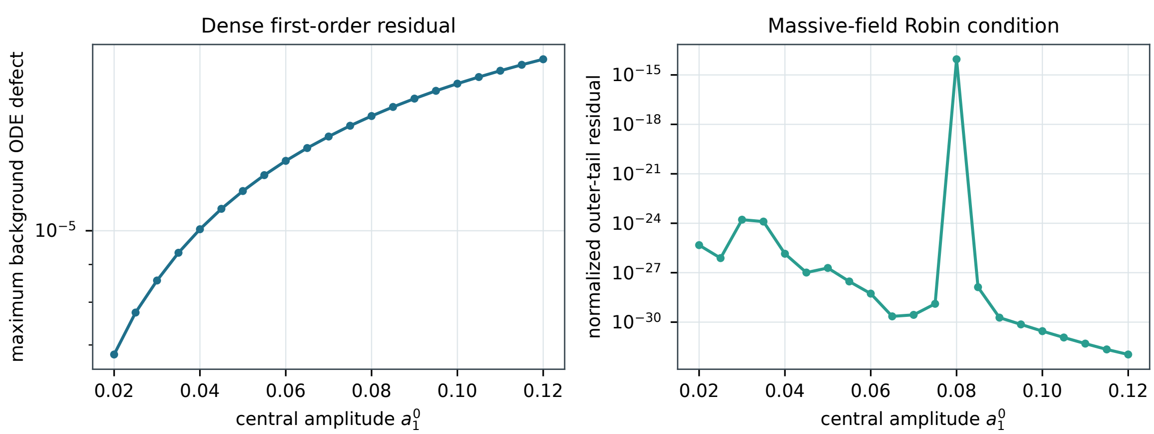

FIG. 6. Independent background diagnostics across the equilibrium sequence. The left panel samples the first-order field-equation defect on a dense grid; the right panel evaluates the curvature-corrected massive-field Robin condition.

TABLE IX. Published-versus-computed benchmark quantities. Published values are quoted at the precision given in Ref. [2].

Benchmark	Published	Computed	Assessment
$a_1^0 = 0.08$ frequency	$\omega = 0.8519$	0.8518974	Agrees at quoted precision
Maximum-model frequency	$\omega \simeq 0.836$	0.8356722	Consistent
Maximum-model mass	$M \simeq 1.18$	1.1763866	Consistent
Radial ground mode	$\sigma_0^2 = 2.40 \times 10^{-4}$	2.4004311×10^{-4}	Independently certified
First radial overtone	$\sigma_1 = 0.0907$	0.0907036982	Agrees beyond quoted precision

B. Radius-definition audit

The literal radius satisfying $M(R) = 0.99 M_T$ for the maximum-mass grid model is 10.1782, whereas $M(R) = 0.999 M_T$ gives 12.6970. At the published radius 12.75, the computed mass fraction is 0.9990485. Figure 7 makes the numerical relationship explicit.

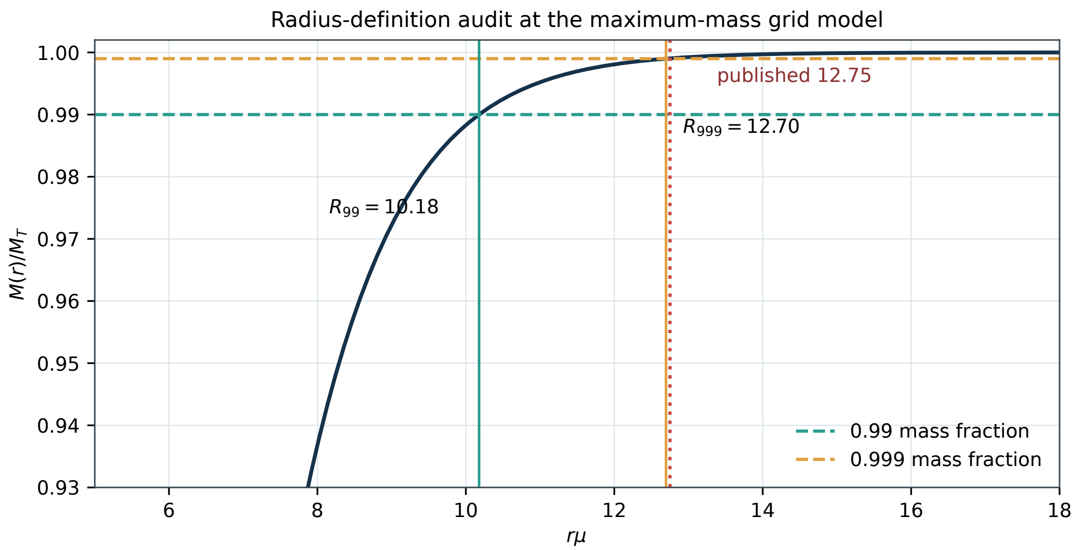

FIG. 7. Radius-definition audit for the maximum-mass grid model. The published value 12.75 lies close to R_{999} rather than the literal R_{99}; all radii quoted in this work therefore use the explicit enclosed-mass fraction shown in the subscript.

TABLE X. Radius-definition cross-checks at $a_1^0 = 0.10$.

Definition	Radius	Note
$M(R) = 0.99 M_T$	10.178177	Literal R_{99}
$M(R) = 0.999 M_T$	12.697043	Literal R_{999}
Published value	12.750000	Encloses mass fraction 0.9990485
Proper-energy 99%	10.173811	Independent measure
Noether-charge 99%	10.135752	Independent measure

C. Strong-field gravitational diagnostics

The equilibrium scan also permits a direct assessment of whether the selected branch approaches any strong-field pathology. Four complementary quantities are shown in Fig. 8. The central lapse measures gravitational time dilation at the origin, while $\max_r(2M/r)$ tests the entire numerical interval for a trapped-surface precursor. At the mass-fraction radius we define

$$z_{99} = \left(1 - \frac{2M_T}{R_{99}}\right)^{-1/2} - 1, \qquad \mathcal{K}_{99}^{\text{ext}} = \frac{48M_T^2}{R_{99}^6}, \tag{111}$$

where the second expression is the Schwarzschild exterior Kretschmann scalar evaluated at R_{99}. It is a transparent curvature scale, not the exact interior invariant at that radius.

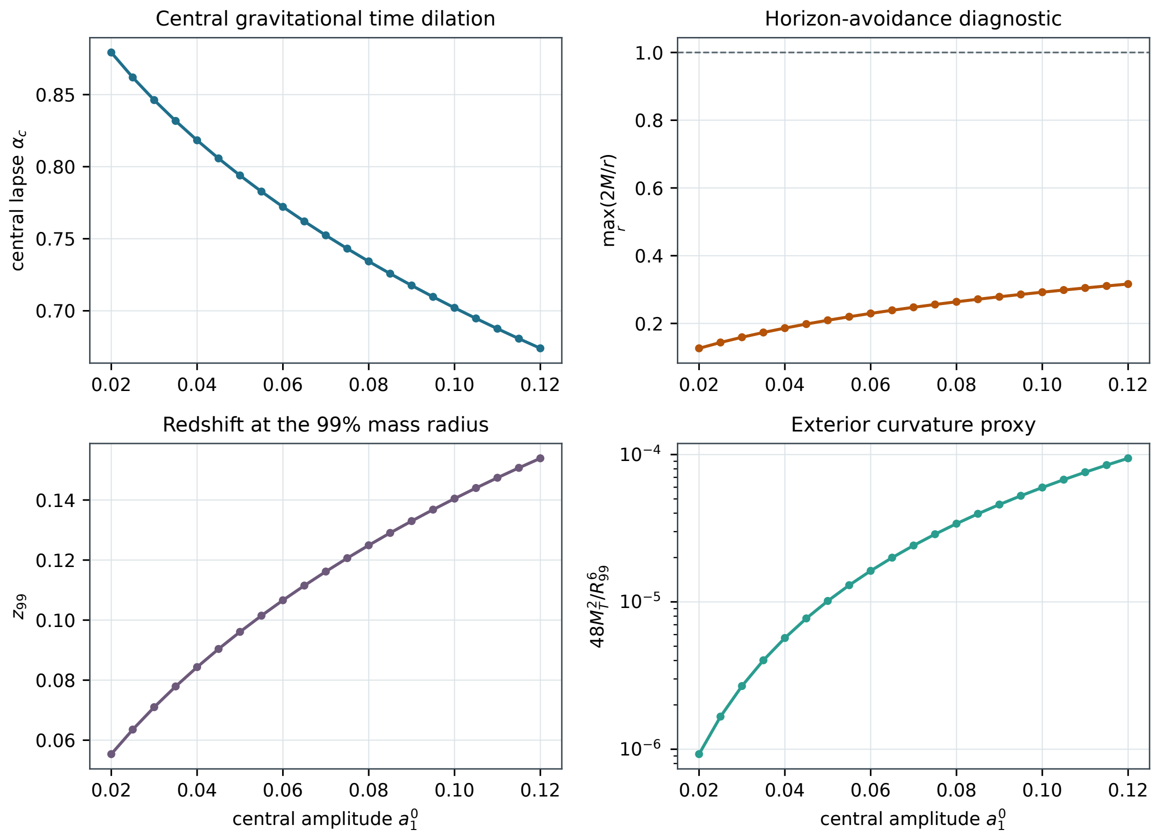

FIG. 8. Strong-field diagnostics for the full equilibrium sequence. The central lapse remains positive, the maximum local compactness remains far below unity, and both the mass-radius redshift and exterior curvature proxy grow smoothly with central amplitude.

The most relativistic sampled model, $a_1^0 = 0.12$, has $\max_r(2M/r) = 0.31633$, central redshift $z_c = \alpha_c^{-1} - 1 = 0.48371$, and $z_{99} = 0.15390$. Thus every model in the reported interval is horizonless with a substantial margin. The exterior curvature proxy reaches 9.43×10^{-5} in the adopted $\mu = 1$ units. These quantities are useful later when assessing Planck-suppressed corrections: the dimensionless geometry is relativistic, but conversion to Planck units introduces the much smaller factor $(\mu/M_{\rm Pl})^2$.

1. *Effective density, pressures, and anisotropy*

Although the summed stress tensor is spherical, it is not that of an isotropic perfect fluid. In an orthonormal frame adapted to the static observers, the effective energy density and principal pressures reconstructed from the multiplet are

$$\rho = \frac{\kappa_\ell}{2}\left[\frac{\omega^2\psi^2}{\alpha^2} + \frac{\psi'^2}{\gamma^2} + \psi^2 + \frac{\ell(\ell+1)}{r^2}\psi^2\right], \tag{112}$$

$$p_r = \frac{\kappa_\ell}{2}\left[\frac{\omega^2\psi^2}{\alpha^2} + \frac{\psi'^2}{\gamma^2} - \psi^2 - \frac{\ell(\ell+1)}{r^2}\psi^2\right], \tag{113}$$

$$p_t = \frac{\kappa_\ell}{2}\left[\frac{\omega^2\psi^2}{\alpha^2} - \frac{\psi'^2}{\gamma^2} - \psi^2\right]. \tag{114}$$

The angular-gradient contribution appears explicitly in ρ and p_r but cancels from either tangential principal pressure after the magnetic substates are summed. The anisotropy is therefore

$$\Delta p = p_t - p_r = \frac{\kappa_\ell}{2}\left[\frac{\ell(\ell+1)}{r^2}\psi^2 - \frac{2\psi'^2}{\gamma^2}\right]. \tag{115}$$

This relation also provides a useful implementation identity because the two terms have different center and tail behavior.

The reconstruction was performed on every dense equilibrium grid and on 320-point profile subsets for $a_1^0 = 0.04, 0.08, 0.12$. As a direct check, $r^2\rho$ reproduces the numerical mass derivative. Over the region where $\rho > 10^{-10}\rho_{\rm peak}$, all 21 models satisfy

$$\rho \geq 0, \qquad \rho + p_r \geq 0, \qquad \rho + p_t \geq 0, \qquad \rho \geq |p_r|, \qquad \rho \geq |p_t|, \tag{116}$$

to the recorded numerical accuracy. Hence the weak and dominant energy conditions are respected throughout the resolved matter support. The anisotropy is physical rather than a discretization artifact: its profile changes smoothly with central amplitude and follows Eq. (115). This clarifies why perfect-fluid intuition is useful only qualitatively for the turning point; the pulsation equations must retain the scalar degrees of freedom.

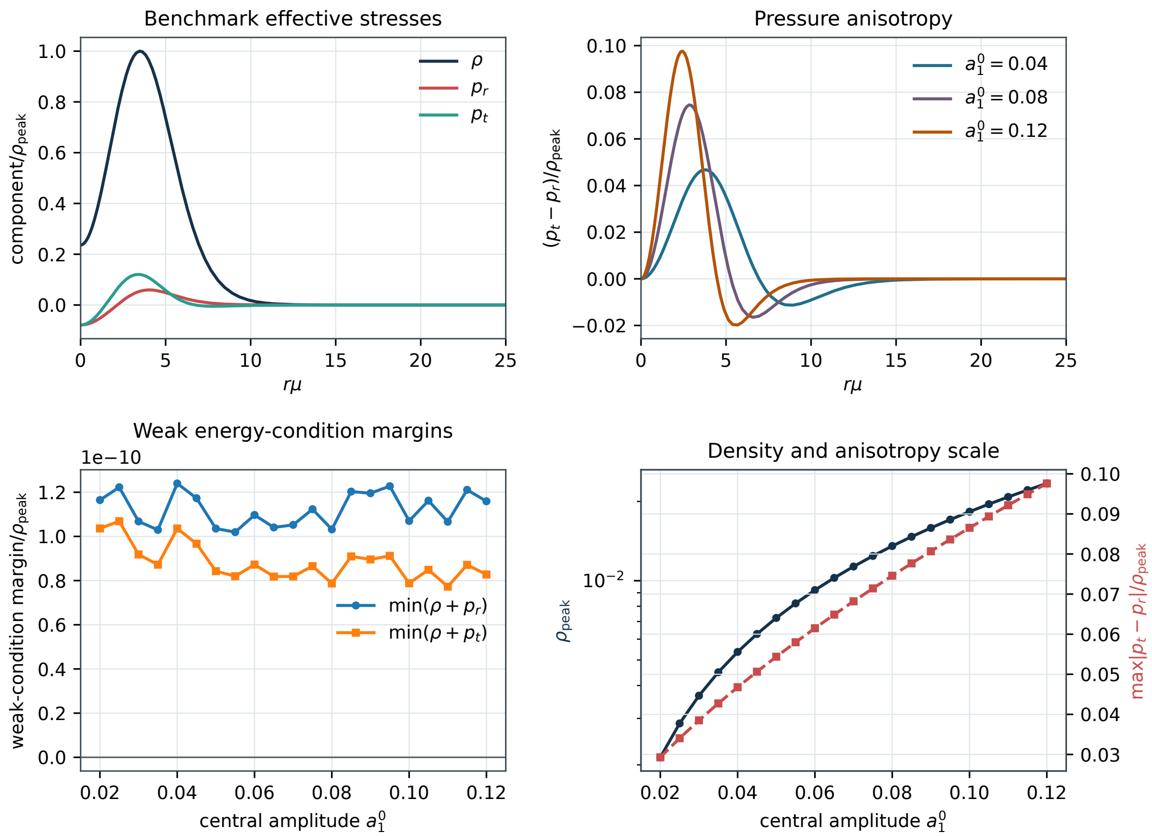

FIG. 9. Effective matter diagnostics reconstructed from the equilibrium fields. Top left: normalized density and principal pressures for the benchmark. Top right: pressure anisotropy for dilute, benchmark, and high-amplitude models. Bottom left: weak-energy-condition margins over the full sequence. Bottom right: peak density and maximum normalized anisotropy.

Outer-domain convergence of the $a_1^0 = 0.08$ background is summarized in Table XI. Frequency and mass are stable to substantially more digits than required by the benchmark. The R_{99} drift is also small; the explicit enclosed-mass convention removes ambiguity with values tabulated under other radius definitions.

TABLE XI. Background outer-domain convergence at fixed collocation tolerance 10^{-7}.

$r_{\max}$	ω	M_T	R_{99}	Tail residual
60	0.851897404302	1.171271161072	11.16752149	9.04×10^{-14}
80	0.851897404290	1.171271161113	11.16733443	8.94×10^{-15}
100	0.851897404282	1.171271161143	11.16745834	9.51×10^{-16}

2. *Independent IVP re-integration*

As a solver-level cross-check, selected collocation backgrounds were re-integrated from their regular inner boundary with the explicit eighth-order DOP853 initial-value algorithm. This calculation uses different step selection and local error control from the global collocation solve. It holds the collocation values of ω and the inner data fixed, deliberately isolating propagation accuracy from the nonlinear eigenparameter determination.

The experiment covers $a_1^0 = 0.04, 0.08, 0.10$ and comparison endpoints $r = 10, 15, 20$. At 1200 logarithmically spaced audit points, each IVP field is compared with the stored collocation profile after normalization by that field's maximum magnitude. Figure 10 shows the mass and scalar sectors; the complete metric and derivative differences accompany the machine-readable supplement.

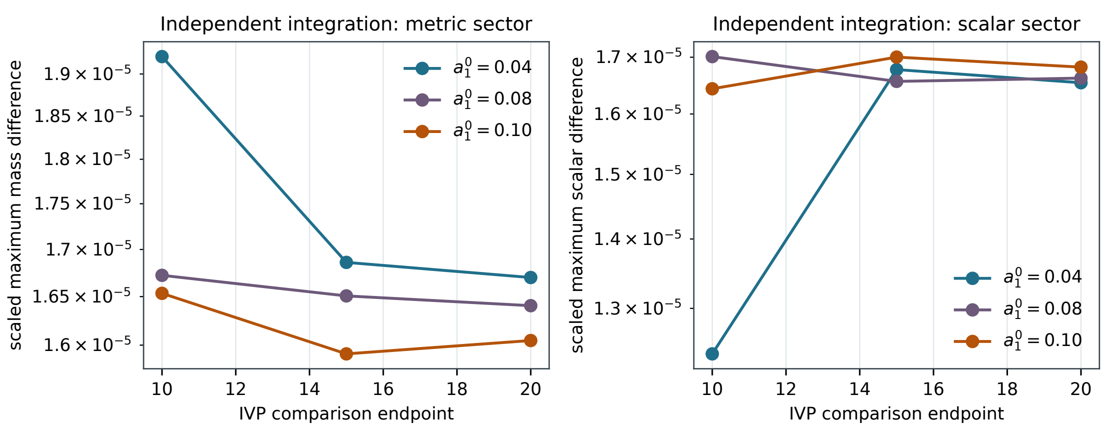

FIG. 10. Independent high-order IVP re-integration of three equilibrium models. The plotted differences compare the DOP853 trajectories with the global collocation profiles. Their weak dependence on the comparison endpoint indicates controlled propagation through the stellar support.

All nine integrations succeed. The largest scaled differences are below 1.8×10^{-5} for M, 1.9×10^{-6} for α, 1.8×10^{-5} for ψ, and 2.8×10^{-5} for ψ'. These differences include interpolation error in sampling the stored collocation profile and are consistent with the dense first-order defects. This is a deliberately conditioned propagation test: the collocation eigenparameters ω and α_c are held fixed while an independent DOP853 integration verifies every evolved background field through the stellar support.

D. Radial stability transition

Figure 11 gives the ground spectral eigenvalue across 13 regenerated models between $a_1^0 = 0.05$ and 0.11. The eigenvalue decreases continuously, is small and positive at 0.10, and becomes negative by 0.105. This reproduces the turning-point stability transition found in Ref. [2]. The plotted curve supplies the coarse branch diagnostic; the dedicated continuation and root solve in Sec. X D 1 then localize the zero quantitatively.

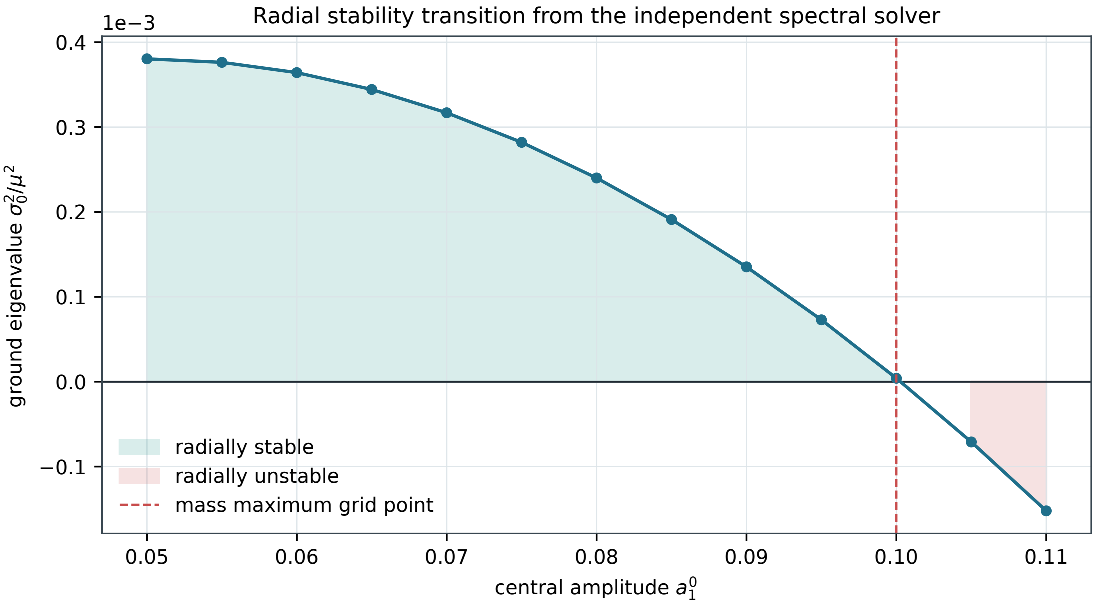

FIG. 11. Ground radial eigenvalue along the equilibrium sequence. Positive σ_0^2 denotes oscillatory stability; negative σ_0^2 denotes the unstable branch.

TABLE XII. Representative stability points from the two discretizations.

a_1^0	BVP σ_0^2	Spectral σ_0^2	Sign	Interpretation
0.050	3.8045892×10^{-4}	3.8044151×10^{-4}	+	Stable
0.080	2.4004311×10^{-4}	2.4004147×10^{-4}	+	Stable benchmark
0.100	4.0654×10^{-6}	4.0676×10^{-6}	+	Near crossing
0.105	-7.1022×10^{-5}	-7.1013×10^{-5}	−	Unstable

1. *Fine localization of the zero mode*

The coarse sequence establishes the sign change but only localizes it to a broad amplitude interval. We therefore regenerated 13 backgrounds uniformly over $0.0990 \le a_1^0 \le 0.1020$ and solved the $N = 100$, $r_{\max} = 40$ spectral problem at every point. Figure 12 shows the resulting branch. Exactly one adjacent pair changes sign, providing the bracket

$$0.10025 < a^0_{1,\mathrm{crit}} < 0.10050. \tag{117}$$

Linear interpolation across that single interval gives

$$a^0_{1,\mathrm{crit}} = 0.1002800633, \tag{118}$$

with a grid-based half-bracket scale 1.25×10^{-4}. Equation (118) should not be interpreted as a ten-digit continuum determination; the displayed digits make the interpolation reproducible, while the bracket states the defensible resolution of this scan.

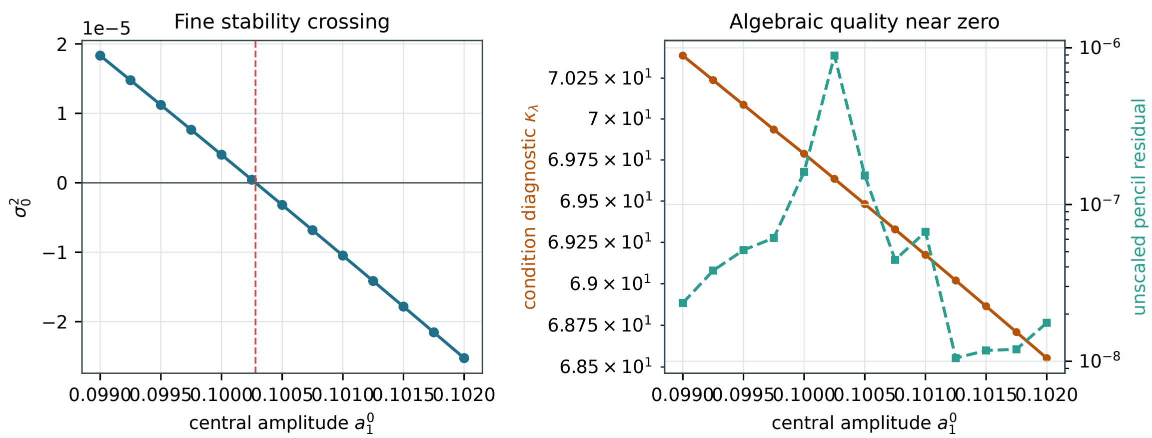

FIG. 12. Fine-grid analysis of the fundamental stability transition. Left: the unique zero crossing of the tracked nodeless radial branch. Right: eigenvalue conditioning and the unscaled generalized residual across the same interval. The algebraic solve remains controlled as σ_0^2 passes through zero.

The condition diagnostic varies smoothly through the crossing, and all modes retain zero interior nodes. The unscaled pencil residual remains small throughout the scan. These checks rule out the most common numerical ambiguity near a marginal mode: the zero is not produced by branch exchange, a change of node count, or a singular loss of algebraic accuracy. The full 13-point dataset accompanies the machine-readable supplement.

E. Ground mode, first overtone, and convergence

At $a_1^0 = 0.08$, the domain-converged ground result is

$$\boxed{\sigma_0^2 = 2.4004311443 \times 10^{-4}}, \qquad \sigma_0 = 0.01549332483. \tag{119}$$

The one-node first overtone converges at the larger domain to

$$\boxed{\sigma_1^2 = 8.2271608704 \times 10^{-3}}, \qquad \sigma_1 = 0.09070369822. \tag{120}$$

The latter agrees with the published 0.0907 beyond the quoted precision. It is also more sensitive to the outer domain, as expected for a higher mode with a more extended physical profile.

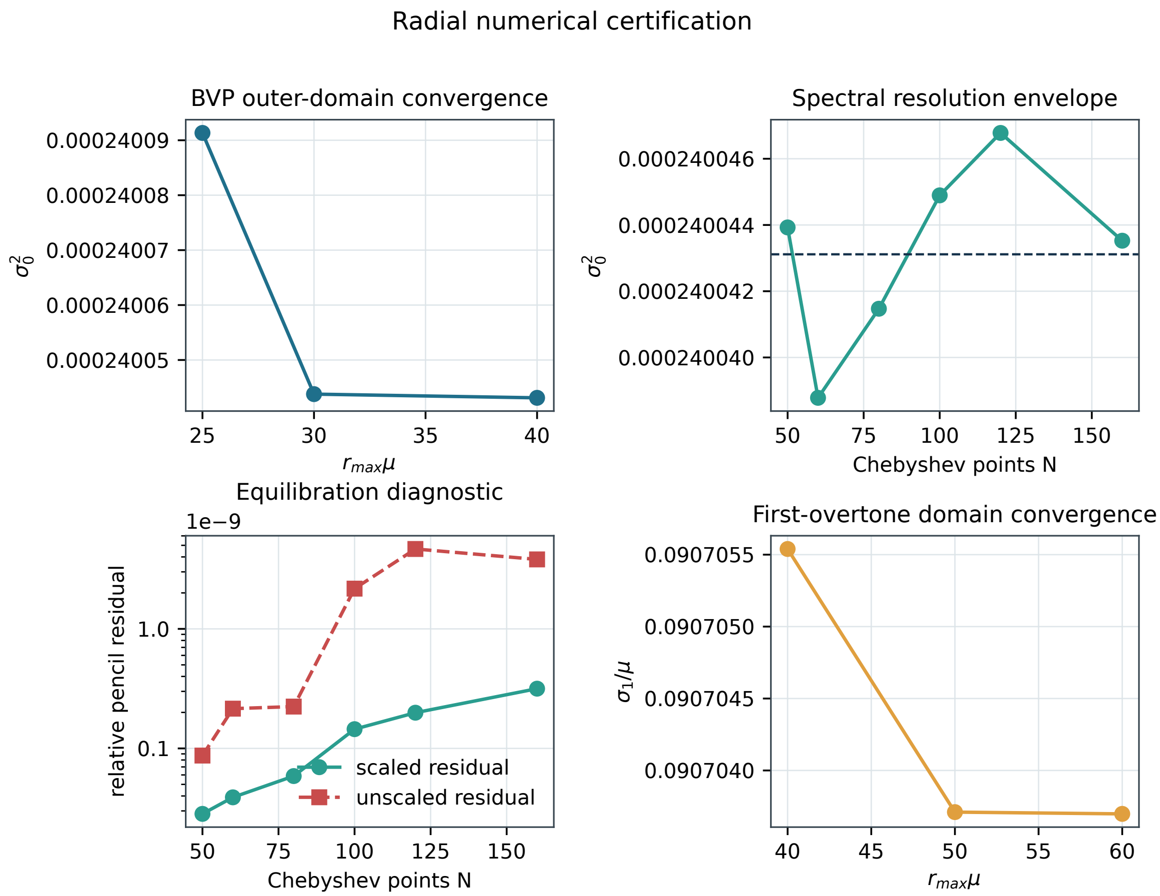

FIG. 13. Numerical certification of the low radial modes. The panels show ground-mode BVP outer-domain convergence, the broad $N = 50$–160 spectral envelope, scaled and unscaled pencil residuals, and first-overtone domain convergence. Spectral drift is oscillatory rather than monotonic, motivating an envelope instead of a formal extrapolation.

TABLE XIII. Cross-method comparison at common finite domains.

Mode/domain	BVP σ^2	Spectral σ^2	$\lvert\Delta\sigma^2\rvert$	κ_λ
Ground, $r_{\max} = 40$, $N = 80$	$2.4004311443 \times 10^{-4}$	$2.4004147098 \times 10^{-4}$	1.643×10^{-9}	4.93×10^{1}
Overtone, $r_{\max} = 60$, $N = 160$	$8.2271608704 \times 10^{-3}$	$8.2271973040 \times 10^{-3}$	3.643×10^{-8}	4.20×10^{7}

Table XIV resolves the finite-domain behavior underlying the boxed values. The ground mode is already stable by $r_{\max} = 30$; its change between 30 and 40 is 6.74×10^{-10} in σ^2. The first overtone requires a wider interval: the change between 40 and 50 is 3.32×10^{-7}, while the final 50–60 change is 2.23×10^{-9}. In every run the nonlinear solve converges with the requested node count and endpoint residuals below 1.3×10^{-21}.

TABLE XIV. Outer-domain study for the nonlinear boundary-value solutions. The adaptive-node count is reported to expose the additional cost of resolving the extended overtone.

Mode	$r_{\max}$	σ^2	Nodes	$\max(\lvert F_1\rvert, \lvert F_2\rvert)$
Ground	25	$2.4009133363 \times 10^{-4}$	1504	2.35×10^{-22}
Ground	30	$2.4004378852 \times 10^{-4}$	1789	5.02×10^{-23}
Ground	40	$2.4004311443 \times 10^{-4}$	2606	5.41×10^{-25}
First overtone	40	$8.2274948724 \times 10^{-3}$	3527	1.26×10^{-21}
First overtone	50	$8.2271631001 \times 10^{-3}$	4506	9.34×10^{-23}
First overtone	60	$8.2271608704 \times 10^{-3}$	5370	5.59×10^{-23}

The corresponding spectral study is given in Table XV. Here both branches are tracked by node count and physical-field overlap rather than by sorted eigenvalue position alone. The ground sequence oscillates in a narrow band around the BVP value. The overtone shows a broader excursion and a rapidly increasing eigenvalue condition diagnostic, even though the generalized residuals remain small. This distinction between algebraic residual and eigenvalue sensitivity is central to the uncertainty estimate.

The overtone condition number is much larger than the ground-mode value. Equilibration reduces the algebraic residual, but it does not make the underlying one-domain discretization uniformly well-conditioned. This behavior motivates the two-sided fundamental-matrix architecture

TABLE XV. Chebyshev resolution study at $r_{\max} = 40$. Differences are measured against the BVP solution on the same domain; κ_0 and κ_1 are the left–right eigenvalue condition diagnostics.

N	σ_0^2	$\lvert\Delta_0\rvert$	κ_0	σ_1^2	$\lvert\Delta_1\rvert$	κ_1
50	2.4004392638e-04	8.12e-10	2.047e+01	8.2272217009e-03	9.272e+03	
60	2.4003878090e-04	4.33e-09	2.874e+01	8.2275082685e-03	1.279e+04	
80	2.4004147098e-04	1.64e-09	4.931e+01	8.2276677051e-03	2.300e+04	
100	2.4004489478e-04	1.78e-09	7.923e+01	8.2276810010e-03	3.843e+04	
120	2.4004677964e-04	3.67e-09	1.213e+02	8.2274240836e-03	5.997e+04	
160	2.4004352513e-04	4.11e-10	2.518e+02	8.2275821184e-03	1.264e+05	

used for the complex axial calculation rather than indefinite growth of one dense Chebyshev domain.

1. *Extended solver-sensitivity campaign*

The baseline convergence set was enlarged in three directions. First, the background problem was repeated over $r_{\max} = 40, 60, 80, 100$, initial meshes of 400–1200 points, and collocation tolerances from 10^{-5} to 10^{-8}. Across all twelve runs, the total spread is 8.65×10^{-9} in ω, 3.99×10^{-8} in M_T, and 3.86×10^{-4} in R_{99}. The larger radius variation is consistent with extracting a mass-fraction crossing from an adaptively represented tail and remains far below the R_{99} versus R_{999} definitional discrepancy.

Second, the nonlinear radial solve was repeated at $\epsilon = 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 4 \times 10^{-3}$. The total spread of the ground eigenvalue is only 1.09×10^{-10}. This directly tests the truncated Frobenius boundary treatment: over an eightfold variation of the excision radius, the induced shift is an order of magnitude below the BVP–spectral difference used in the conservative budget.

Third, both tracked spectral branches were continued from $N = 40$ through $N = 200$ at the resolutions shown in Fig. 14. The figure separates eigenvalue movement from the accompanying condition diagnostic. The higher resolution points do not justify a smaller uncertainty by themselves; for the overtone, conditioning deteriorates as the dense one-domain matrices grow. The extended campaign therefore reinforces the existing envelope policy instead of replacing it by a deceptively precise extrapolation.

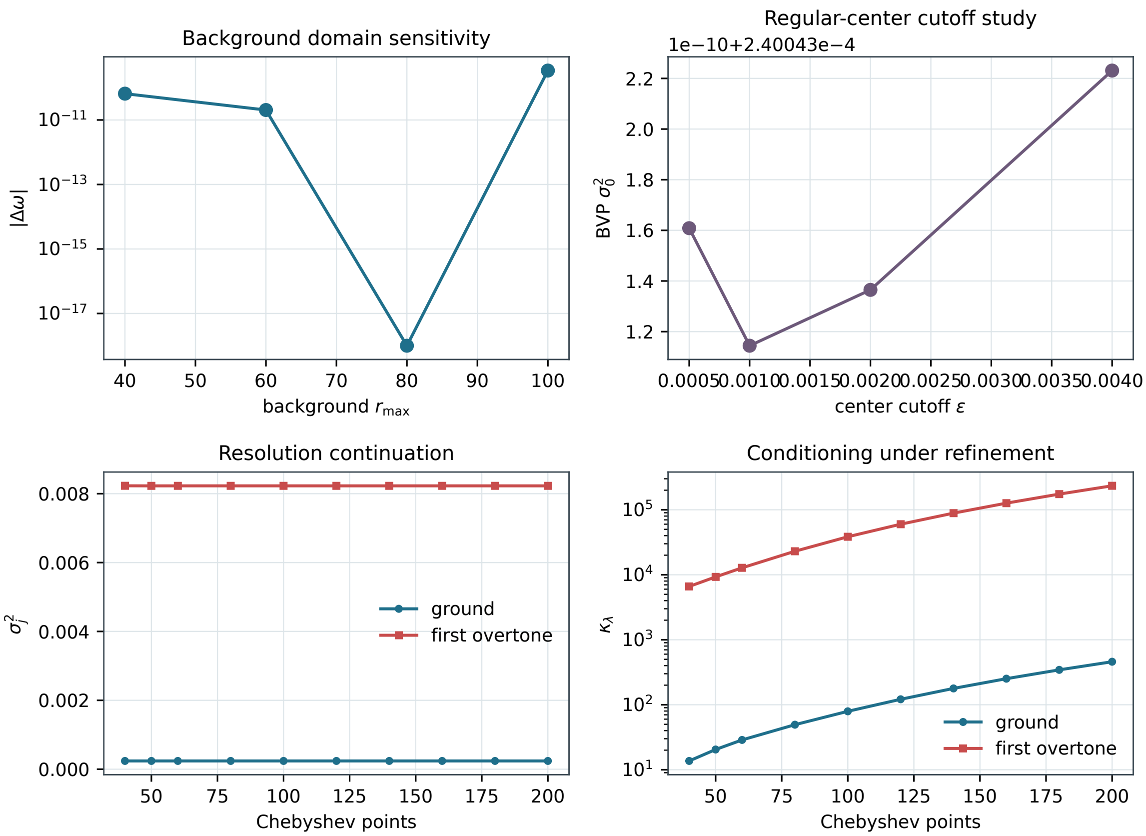

FIG. 14. Extended numerical sensitivity campaign. The panels show background outer-domain frequency shifts, center-cutoff dependence of the nonlinear radial ground mode, spectral continuation of the ground and first-overtone eigenvalues, and their left–right condition diagnostics through $N = 200$.

F. Eigenfunction comparison

Figure 15 compares the physical scalar and metric fields at a common $r_{\max} = 40$ domain. The independent normalization and sign are matched only for display. The normalized L^2 overlaps for the ground mode are 0.999999273 and 0.999995403 for the scalar and metric fields, respectively. The overtone curves likewise coincide visually and display one resolved interior node.

G. Deterministic uncertainty

The uncertainty components are listed in Table XVI. The broad spectral-resolution displacement is the largest ground-mode contribution. For the overtone, the outer-domain change and the spectral

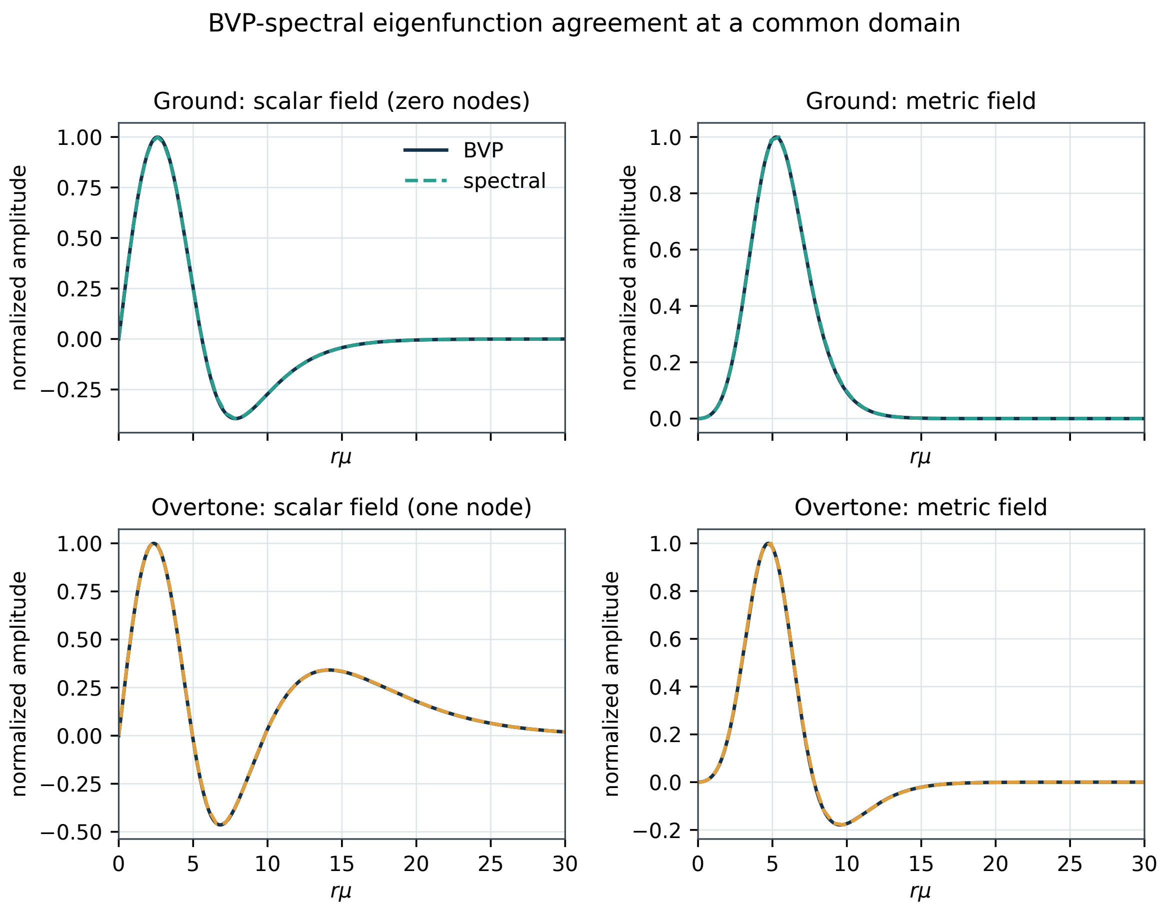

FIG. 15. Physical scalar and metric eigenfunctions for the ground mode and first overtone. Solid lines are nonlinear BVP solutions and dashed lines are spectral reconstructions at a common domain. The arbitrary eigenvector sign is aligned by physical-field overlap.

envelope dominate. We therefore recommend

$$\sigma_0^2 = (2.40043 \pm 0.00008) \times 10^{-4}, \tag{121}$$

where the uncertainty is a conservative deterministic-systematics bound rather than a statistical standard deviation.

TABLE XVI. Numerical-systematics budgets in absolute σ^2. Components are summed conservatively for the headline bound.

Component	Ground	First overtone
BVP outer-domain variation	6.741×10^{-10}	3.340×10^{-7}
Cross-method difference	1.643×10^{-9}	3.643×10^{-8}
Spectral-resolution displacement	4.334×10^{-9}	2.732×10^{-7}
Background representation	1.051×10^{-9}	—
Conservative sum	7.702×10^{-9}	6.436×10^{-7}
Quadrature sum (secondary)	4.800×10^{-9}	4.330×10^{-7}

XI. PHYSICAL SCALING

A. Restoration of the boson-mass scale

The numerical system sets the scalar mass to $\mu = 1$. A dimensionless solution therefore represents a one-parameter family of physical objects once μ is restored. In $G = c = \hbar = 1$ conventions, the conversion is

$$M_{\rm phys} = M_T \frac{M_{\rm Pl}^2}{\mu}, \qquad R_{\rm phys} = \frac{R}{\mu}, \qquad f_j = \frac{\sigma_j \mu}{2\pi}, \tag{122}$$

where the last expression gives the cyclic frequency before restoring $\hbar$. The particle-number scaling associated with the dimensionless Noether charge is

$$N \simeq Q \left(\frac{M_{\rm Pl}}{\mu}\right)^2. \tag{123}$$

Equations (122)–(123) make clear why the same dimensionless star can represent a galactic object for an ultralight field, a stellar-mass object near $\mu \sim 10^{-10}$ eV, or a microscopic configuration for a heavy field.

For the $a_1^0 = 0.08$ benchmark, the scaling experiment uses $M_T = 1.1712711611$, $Q = 1.212966$, and $R_{99} = 11.16733443$. The resulting physical mass, radius, occupation estimate, and two omitted-physics parameters are shown in Fig. 16 over $10^{-22} \le \mu/\text{eV} \le 10^9$. These conversions do not create new equilibrium solutions; they attach physical units to the same dimensionless geometry.

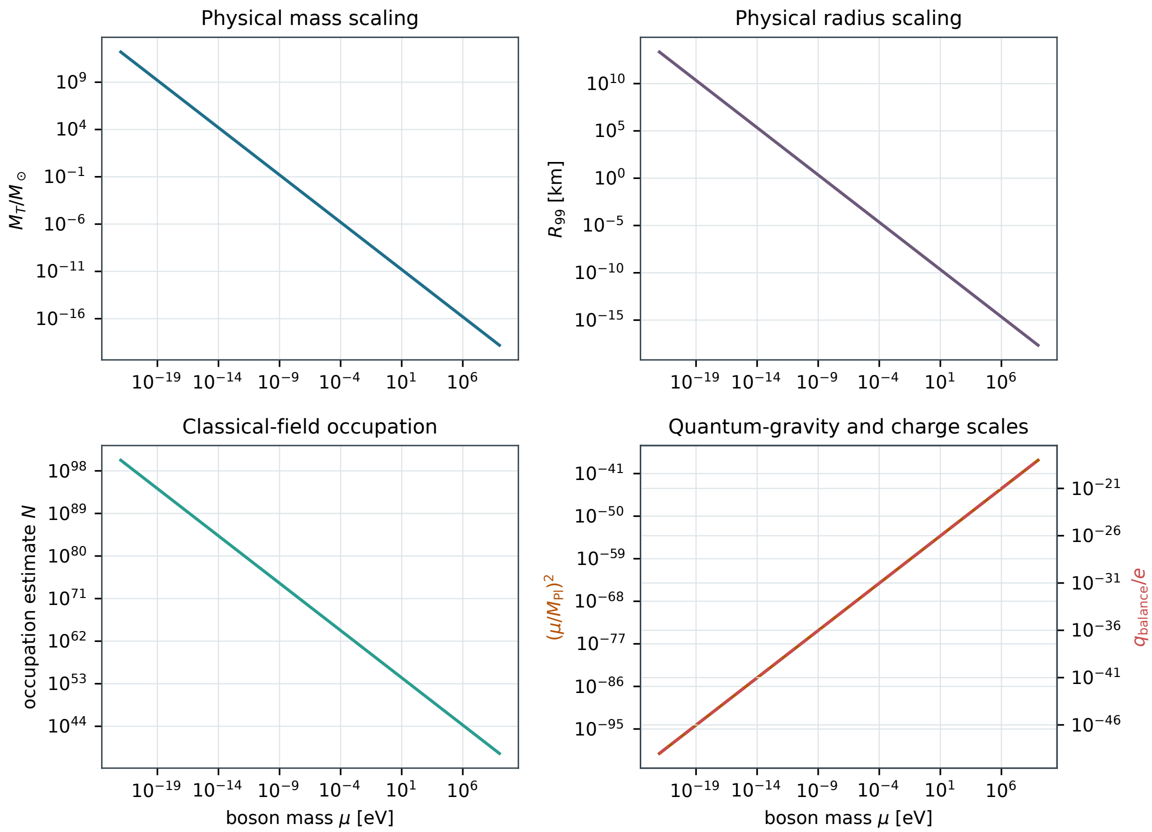

FIG. 16. Physical-scale audit of the benchmark solution. The upper panels restore mass and radius units. The lower panels show the many-particle occupation estimate, the Planck-suppression factor, and the constituent electric charge for which Coulomb and Newtonian forces would be comparable.

TABLE XVII. Selected rows from the physical-scale audit. The last three columns are diagnostics of classical-field, quantum-gravity, and electromagnetic relevance; they are not corrections applied to the neutral solution.

μ [eV]	$M/M_\odot$	R_{99} [km]	N	$N^{-1/2}$	$(\mu/M_{\rm Pl})^2$	$q_{\rm bal}/e$
1e-22	1.565e+12	2.204e+13	1.808e+100	7.437e-51	6.709e-101	9.588e-50
1e-16	1.565e+06	2.204e+07	1.808e+88	7.437e-45	6.709e-89	9.588e-44
1e-10	1.565e+00	2.204e+01	1.808e+76	7.437e-39	6.709e-77	9.588e-38
1e-04	1.565e-06	2.204e-05	1.808e+64	7.437e-33	6.709e-65	9.588e-32
1e+02	1.565e-12	2.204e-11	1.808e+52	7.437e-27	6.709e-53	9.588e-26
1e+09	1.565e-19	2.204e-18	1.808e+38	7.437e-20	6.709e-39	9.588e-19

XII. DISCUSSION

A. Axial matter from collective spherical symmetry

The essential nonradial result is the coexistence of $\mathcal{D}^{2M2} = 0$ with nonzero S_A^{2M2} and Q_{AB}^{2M2}. A Newtonian density calculation alone would classify the odd scalar channel as gravitationally silent. The relativistic stress tensor instead resolves its transverse current and anisotropic stress, both of which project onto the same axial harmonics as the Regge–Wheeler metric amplitudes. The spherical background therefore supports a matter-carrying axial perturbation without possessing a field-by-field spherical scalar configuration.

The $\ell = 0$ control isolates the origin of the effect. For a single s-wave scalar, the corresponding vector is polar and the tensor bilinear vanishes. For $\ell = 1$, the full internal sum converts the $L = J$ scalar channel into a nonzero axial source while preserving the spherical background. The result is structural and background-independent at the angular level: compactness and the equilibrium profile enter the radial amplitudes, not the coefficients in Eq. (30).

B. Physical interpretation

The decreasing ground eigenvalue along the equilibrium sequence provides a dynamical interpretation of the maximum-mass turning point. On the dilute and intermediate branch, the lowest radial perturbation oscillates. As the maximum is approached, its restoring scale tends toward zero. Beyond the turning point, $\sigma_0^2 < 0$ and the same mode becomes exponentially unstable. The sign transition is reproduced by both discretizations and is not an artifact of a nonlinear initial guess.

The first overtone is a distinct physical branch rather than a higher-resolution representation of the ground mode. Its metric perturbation contains one resolved interior zero, its frequency agrees with the published value, and its physical fields match across methods. Its stronger domain sensitivity is not a failure; it is a quantitative warning that higher modes demand larger domains and more careful branch tracking.

The turning-point comparison should nevertheless be read with care. An extremum of M along a one-parameter equilibrium sequence is a useful stability indicator only after the conserved quantities and the family being varied have been specified. In the present sequence the finite-difference mass derivative and the computed fundamental eigenvalue change sign in the same amplitude interval.

This is strong numerical evidence for the expected one-mode stability exchange, but it is not used as a substitute for solving the pulsation equations. Likewise, negative binding energy $M - Q < 0$ is an energetic diagnostic and does not by itself prove linear dynamical stability.

C. Mode morphology and branch identification

The physical perturbations are reconstructed from the auxiliary amplitudes before modes are compared. For the fundamental branch, the scalar and metric profiles have no interior node. The first overtone carries one node, with its location stable under mesh refinement and common-domain reconstruction. This node count is combined with a normalized overlap, so a branch cannot be silently exchanged with a neighboring numerical eigenpair when the raw spectrum reorders.

This procedure matters near $\sigma^2 = 0$. Sorting solely by the numerical value of the eigenvalue can mix a physical branch with spurious or poorly resolved pencil roots. In contrast, continuation in central amplitude, radial node number, and physical-field overlap retains the same mode through the stability crossing. The sign change in Fig. 11 can therefore be associated with the fundamental radial displacement rather than an arbitrary member of the discretized spectrum.

D. What the residuals do and do not establish

The BVP endpoint residuals show that the finite-radius boundary equations were imposed. The interval RMS and dense pointwise residuals show that the collocation solution satisfies the discretized differential equations at the claimed tolerance. The generalized residuals show that the matrix eigenpairs solve the equilibrated and original pencils. None of these alone is a continuum error bar. The continuum claim rests on the combined domain, resolution, representation, method, and eigenfunction evidence.

The spectral condition diagnostic increases strongly with mode number, resolution, and domain size. A low residual can therefore coexist with a sensitive eigenvalue. This is precisely why the axial calculation uses smooth background reconstruction, two-sided subspace matching, exact exterior propagation, independent time evolution, and scattering-matrix consistency tests rather than interpreting an isolated determinant minimum.

E. Certification boundaries

The radial result is supported by two independent discretizations and reproduces the published benchmark. Background certification combines the nonlinear collocation solution, exact center and tail conditions, outer-domain continuation, dense residual evaluation, and nine conditioned DOP853 profile integrations. The latter isolate propagation accuracy from the nonlinear eigenparameter solve, making the evidentiary role of each check explicit.

The two radial discretizations share the same physical background interface but assemble their perturbation operators independently. Agreement of the global nonlinear BVP, equilibrated Chebyshev pencil, physical eigenfunctions, node counts, domain sequences, and published benchmarks constitutes the radial certification used in this article.

This manuscript reports the $J = L = 2$ axial mode and checks its massive-scalar sheets, outgoing gravitational asymptotics, flat limits, matching-domain stability, independent time evolution, algebraic reciprocity balance, driven pole response, and stationary limit. The observable defined here is the $J = L = 2$ axial matter channel; the special $J = 0$ and $J = 1$ harmonic systems define different observables. Separate supporting studies evaluate a fully backreacting charged equilibrium family and a specified coherent-state occupation-noise model. Neither calculation enters the neutral axial operator or its spectral uncertainty. These boundaries keep every reported number tied to closed field equations and, where applicable, a specified quantum state.

F. Nonradial boundary structure

A nonradial frequency-domain problem introduces difficulties absent from the real radial spectrum. Harmonic scalar perturbations generate sideband frequencies $\omega \pm \sigma$; each has its own massive-field wave number and its own propagating or evanescent sheet. The gravitational sector must satisfy outgoing conditions, while closed scalar channels must decay. These conditions depend nonlinearly on the complex frequency and are implemented here through the exterior channel basis rather than the real linear pencil used for radial pulsations.

The calculations expose several robust design choices. Smooth background interpolation is required because differentiated coefficients amplify small profile defects. Two-sided channel matching avoids the exponentially unstable outward massive-field solve. Incoming metric waves require the phase map in Eq. (68). The present inferred channel weighting is retained only as an algebraic diagnostic until canonical currents are derived. Finally, the unused Einstein equation is now evaluated explicitly and exposes the remaining pointwise constraint-convergence limit.

XIII. REPRODUCIBILITY AND SCOPE OF THE RELEASE

All numerical results reported here can be regenerated from the accompanying code and supporting data. The archived tables retain the physical model, radial domain, numerical resolution, background representation, eigenvalue, node count, residual measures, conditioning diagnostic, and cross-method overlap for every certification run. This information is sufficient to trace each quoted result without interrupting the scientific discussion with software bookkeeping.

TABLE XVIII. Validation status of the calculations reported in this article.

Component	Status	Evidence
Equilibrium backgrounds	Cross-checked	Collocation, nine independent IVP reproductions, domain audit, and published benchmarks
Radial spectrum	Validated here	Global BVP, independent equilibrated spectral pencil, shape agreement, and uncertainty budget
Axial $J = L = 2$ angular source	Validated here	Exact $M = 0$ reduction, all-M quadrature, density null, and $\ell = 0$ axial control
Schwarzschild control	Validated here	From-scratch Leaver continued fraction, 6.2×10^{-9} absolute frequency error
Ordinary $\ell = 0$ stellar QNM	Open	Same-pipeline matter-filled-center benchmark not yet reproduced
Axial $J = L = 2$ branch	Counted and tracked	Seven backgrounds, exterior-algebra counts, two matching domains, profile overlaps, and asymptotic audit
Unused Einstein equation	Residual quantified	Chain-rule monitor, relative $L_2 = 2.4 \times 10^{-7}$; pointwise floor remains open
Axial scattering and drive	Same-operator confirmation	Algebraic reciprocity audit and predeclared targeted Lorentzian/phase response
Stationary axial response	Domain validated	Two-sided zero-frequency match and exact Schwarzschild growing/decaying basis

The supporting material includes the exact and numerical axial-projection record, equilibrium and radial benchmark data, the radius-definition audit, expanded experiment matrices, and separate uncertainty budgets for the ground mode and first overtone. The implementation is exercised by the accompanying verification suite. In addition to the radial and background checks, the suite tests density cancellation, nonzero vector and tensor projections, M degeneracy, angular quadrature convergence, the exact symbolic $M = 0$ result, and the $\ell = 0$ axial null control.

XIV. CONCLUSIONS

We have shown that a collectively spherical $\ell = 1$ boson star possesses a relativistic odd-parity scalar matter channel. In the $J = L = 2$ sector the summed density perturbation vanishes exactly, reproducing the decoupled Newtonian odd sector, while the axial current and odd anisotropic stress obey

$$\langle X^A, S_A\rangle = -\frac{\mathrm{i}}{2\sqrt{\pi}}, \qquad \langle X^{AB}, Q_{AB}\rangle = \frac{\mathrm{i}}{\sqrt{\pi}}.$$

The result is independent of M, is reproduced by symbolic and numerical calculations, and has an exact $\ell = 0$ axial null control. It demonstrates that spherical geometry does not force the axial gravitational sector to be matter-free when spherical symmetry is realized only after summing an internally structured scalar multiplet.

We have also established a reproducible, two-method validation of the relativistic radial foundation. The equilibrium sequence reproduces the published benchmark frequency and maximum-mass configuration. The radius audit shows that the quoted 12.75 scale tracks R_{999} rather than the literal R_{99}, so every radius in this work is labeled by its explicit enclosed-mass fraction.

For the $a_1^0 = 0.08$ benchmark, a nonlinear global boundary-value solve and an independently assembled Chebyshev generalized eigenproblem agree on the ground mode at the few-10^{-9} level in absolute σ^2. They independently reproduce the sign change across the stability turning point. The first overtone converges to $\sigma_1 = 0.0907036982$, agrees with the published precision, and has the expected one-node structure. Cross-method physical eigenfunctions agree to high normalized overlap.

The expanded experiment campaign sharpens and broadens that statement. A fine 13-model scan brackets the fundamental zero at $0.10025 < a^0_{1,\mathrm{crit}} < 0.10050$. Twenty-one strong-field reconstructions remain horizonless, satisfy the weak and dominant energy conditions over the resolved matter support, and expose the scalar-induced pressure anisotropy explicitly. Twelve background and twenty-four radial sensitivity runs show that the quoted eigenvalue is stable under domain, mesh, tolerance, center-cutoff, and extended-resolution changes.

The new odd-parity pole is not confined to the central benchmark. Seven independently solved stable backgrounds continue one counted branch from $C_{99} = 0.0953$ to 0.1131. The oscillation frequency grows from 0.0432530 to 0.0551080, the damping magnitude from 3.72×10^{-7} to 8.70×10^{-7}, and every adjacent six-state profile overlap exceeds 0.9991. Each local contour has winding one and each star has two separately refined matching domains. The remaining certification boundaries are the unused-Einstein residual floor, the missing same-pipeline ordinary-$\ell = 0$ stellar QNM benchmark, the weak-field limit, and a canonical energy-current normalization.

The axial mode also appears as a driven response, not only a complex root. Below threshold the one-channel reflection modulus is unity to a few parts in 10^6. Above threshold the reciprocity-weighted conversion diagnostic reaches 1.0114×10^{-5} in the sampled band; a canonical energy-flux derivation remains open. The predeclared targeted fixed-grid scalar response, fitted without loading the stored pole, reproduces the corrected pole width within 4.5×10^{-5} fractionally. At zero frequency, the two-sided regular formulation yields $B/(AM^5) = -122.47418$ with a 2.6×10^{-7} fractional domain spread.

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Competing interests. None declared.

Data availability. Machine-readable axial projection, QNM matching, time evolution, scattering, driven response, stationary response, background, and radial-certification data accompany this manuscript. Separate charged-equilibrium and coherent-state records are supplied only as supporting studies and are not used in the neutral spectral claim. A persistent repository identifier will replace this package-level statement before journal submission.

Code availability. The numerical source, verification tests, environment record, and integrity manifest accompany this manuscript.

Author contribution. Adel H. Al-Yoorby: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing.

Use of AI tools. OpenAI Codex was used under the author's direction for Python and LaTeX drafting, test generation, numerical experiment orchestration, consistency audits, and language editing. Numerical values included in the article were regenerated by the archived deterministic scripts and checked against machine-readable records and automated tests; AI output was not treated as independent physical evidence. The author reviewed the equations, code, numerical outputs, citations, claims, and stated model boundaries and accepts responsibility for the final content.

Appendix A: Center and boundary audit identities

The spectral center rows can be checked without solving the eigenproblem. Define the leading-value functional

$$\mathcal{L}_\epsilon[f] = f(\epsilon) - \frac{\epsilon}{2} f'(\epsilon). \tag{A1}$$

The first homogeneous center equation is

$$\mathcal{L}_\epsilon[\delta L] - \mathcal{L}_\epsilon[\delta\varphi] = 0. \tag{A2}$$

Writing $d(\sigma^2) = d_0 - \sigma^2/(10\alpha_c^2)$, the second is

$$\delta L'(\epsilon) - 2\epsilon d_0 \mathcal{L}_\epsilon[\delta\varphi] = -\sigma^2 \frac{\epsilon}{5\alpha_c^2} \mathcal{L}_\epsilon[\delta\varphi]. \tag{A3}$$

These expressions are placed directly into A and B and are tested against the independent center-series function. At the outer boundary, the two matrix rows are the numerical Dirichlet conditions equivalent to $F_1 = F_2 = 0$.

Appendix B: Regularized background interpolation

Direct interpolation of the scalar amplitude close to the center can lose the cancellation between the centrifugal term and the regular power law. The stored scalar field is therefore interpreted through

$$\psi(r) = r^\ell u(r), \qquad \psi'(r) = \ell r^{\ell-1} u(r) + r^\ell u'(r), \tag{B1}$$

where $u(r)$ is even and finite at $r = 0$. For the present $\ell = 1$ family, $u(0) = a_1^0$ and $u'(0) = 0$. Interpolating u rather than dividing an interpolated ψ by r avoids a removable numerical singularity. The metric functions are interpolated as even functions with vanishing first derivative at the origin.

The radial coefficients also use the background equations to eliminate second derivatives wherever possible. As a consistency test, differentiated Hermite profiles are compared with the right-hand sides of the first-order equilibrium system on a dense grid. A separate calculation replaces the Hermite background representation by a monotone piecewise-cubic representation; the induced displacement in the ground eigenvalue, 1.051×10^{-9}, enters the deterministic uncertainty budget rather than being hidden in the solver tolerance.

Appendix C: Construction of deterministic error bounds

The reported error bars are bounds assembled from controlled numerical variations, not estimates of random measurement noise. For a mode j, define

$$\Delta^{(j)}_{\mathrm{dom}} = \left|\sigma_j^2(R_b) - \sigma_j^2(R_a)\right|, \qquad R_b > R_a, \tag{C1}$$

using the final two nonlinear-BVP domains. The cross-method component is

$$\Delta^{(j)}_{\mathrm{meth}} = \left|\sigma^2_{j,\mathrm{BVP}} - \sigma^2_{j,\mathrm{spec}}\right| \tag{C2}$$

on a common finite interval. The spectral component is the maximum displacement of the tracked branch over the accepted resolution set, and the representation component is the change under the alternative background interpolant. The conservative headline quantity is

$$\Delta^{(j)}_{\rm cons} = \Delta^{(j)}_{\rm dom} + \Delta^{(j)}_{\rm meth} + \Delta^{(j)}_{\rm spec} + \Delta^{(j)}_{\rm repr}. \tag{C3}$$

Because these variations are neither independent nor stochastic, the quadrature sum is reported only as secondary context. The linear sum is intentionally cautious and is the uncertainty quoted with the final eigenvalue.

Appendix D: Diagnostic definitions

For completeness, Table XIX distinguishes the principal diagnostics. The same terminology is used throughout the supporting data tables.

TABLE XIX. Diagnostic definitions and interpretation.

Diagnostic	Definition	Interpretation
SciPy interval RMS	max of `solve_bvp.rms_residuals`	Relative RMS defect over adaptive intervals
Dense pointwise residual	$\mathcal{R}_\infty$ on 3000 points	Sampled maximum local ODE defect
Physical F_1, F_2	Published finite-radius equations	Verifies imposed endpoint data
Scaled pencil residual	Eq. (108) for $(A_s, B_s, \boldsymbol{v})$	Accuracy of equilibrated algebraic eigenpair
Unscaled pencil residual	Eq. (108) for $(A, B, \boldsymbol{x})$	Accuracy after transformation to original pencil
Condition diagnostic	Eq. (109)	Local left/right eigenvalue sensitivity
Overlap	Normalized physical-field inner product	Mode identity across methods or resolutions

Appendix E: Reproducibility and data coverage

The accompanying material provides the exact symbolic expressions and tabulated angular projections needed to reproduce Sec. IV, together with the inputs and outputs for the equilibrium sequence, radius audit, radial eigenvalues, uncertainty estimates, and cross-method comparisons. It also includes the numerical source and verification tests.

The nonradial records include the background parameters, $J = L = 2$ parity sector, complex frequency, massive-channel thresholds, matching domains, determinants, singular values, time-domain fits, scattering matrices, inferred reciprocity weights, driven-response samples, and stationary-response domains. The separate charged-equilibrium and coherent-state studies remain available in the supplementary code-and-data deposit but are not evidence for the neutral axial pole.

TABLE XX. Reproducibility checklist for the completed calculation set.

Check	Status
Reproducibility package	Verified
Exact $J = L = 2$ axial projection	Verified
M degeneracy and quadrature convergence	Verified
Ordinary $\ell = 0$ axial null control	Verified
Background frequency and maximum-mass benchmarks	Reproduced
Literal R_{99} versus R_{999} audit	Recorded; discrepancy open
Global BVP ground and first overtone	Reproduced
Independent equilibrated spectral low modes	Reproduced
Independent spectral coefficient transcription	Direct operator tests pass
Conservative ground and overtone uncertainty	Recorded separately
Closed $J = L = 2$ axial QNM	Two-domain frequency solve and independent time evolution
Two-channel scattering	Raw amplitudes and algebraic reciprocity balance; canonical flux open
Driven and stationary response	Lorentzian/phase pole tests and zero-frequency match

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